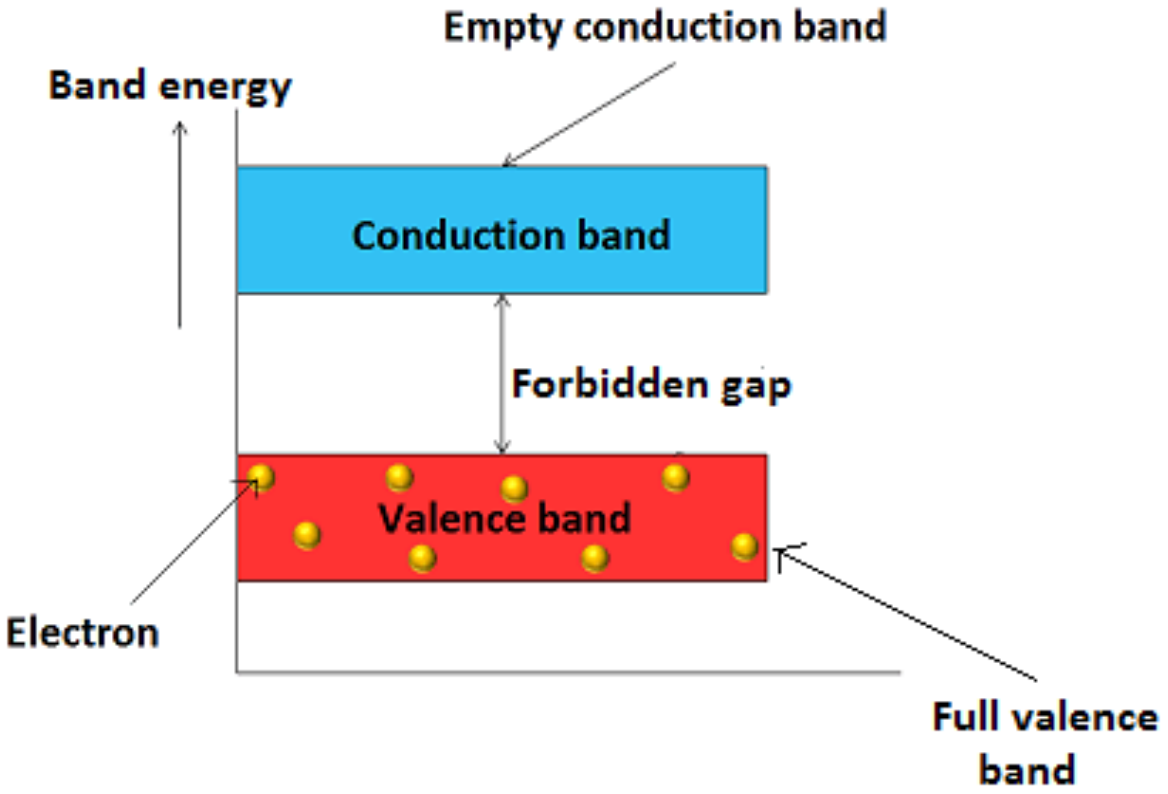
Semiconductors Fundamentals

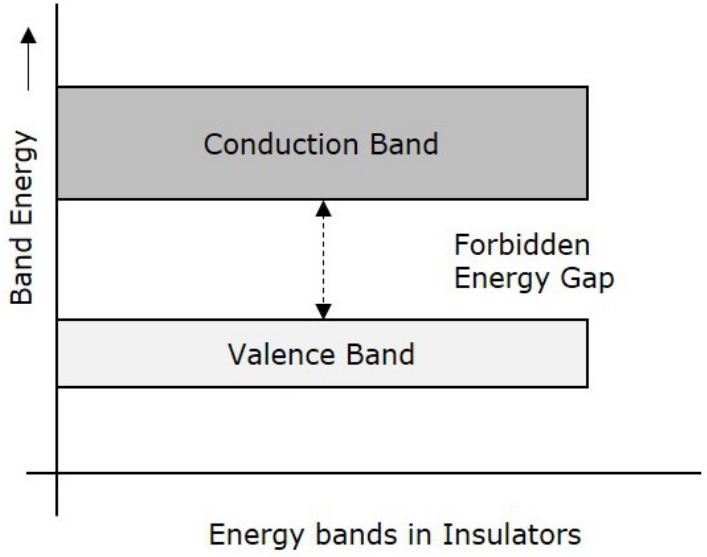
The following subjects will be covered

- Semiconductors in General.
- The PN Junction.
- The Energy Band Diagram.
- Carrier Statistics in Semiconductors.
- PN Junction Band Diagram.
- Photon Generation and Absorption.
- Direct and Indirect Gap Materials.
- Examples.

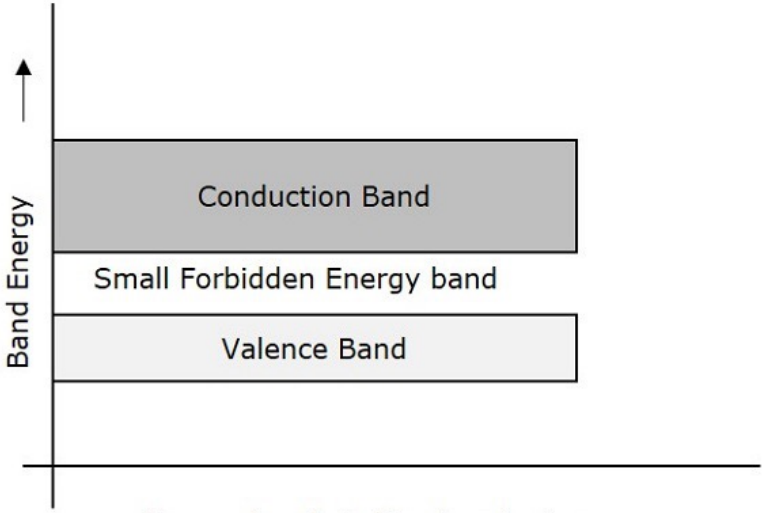
Energy Bands: Isolated atoms and Crystals



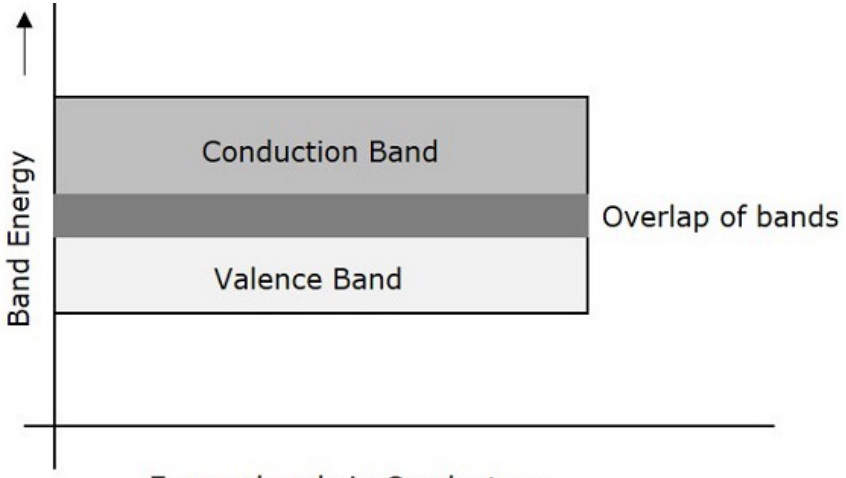
Classification of Materials



Energy bands in Insulators



Energy bands in Semiconductors



Energy bands in Conductors

Examples

Quartz	E_g= 10 eV
Silicon	E_g= 1.12 eV
Copper	E_g= 0 eV

Semiconductors

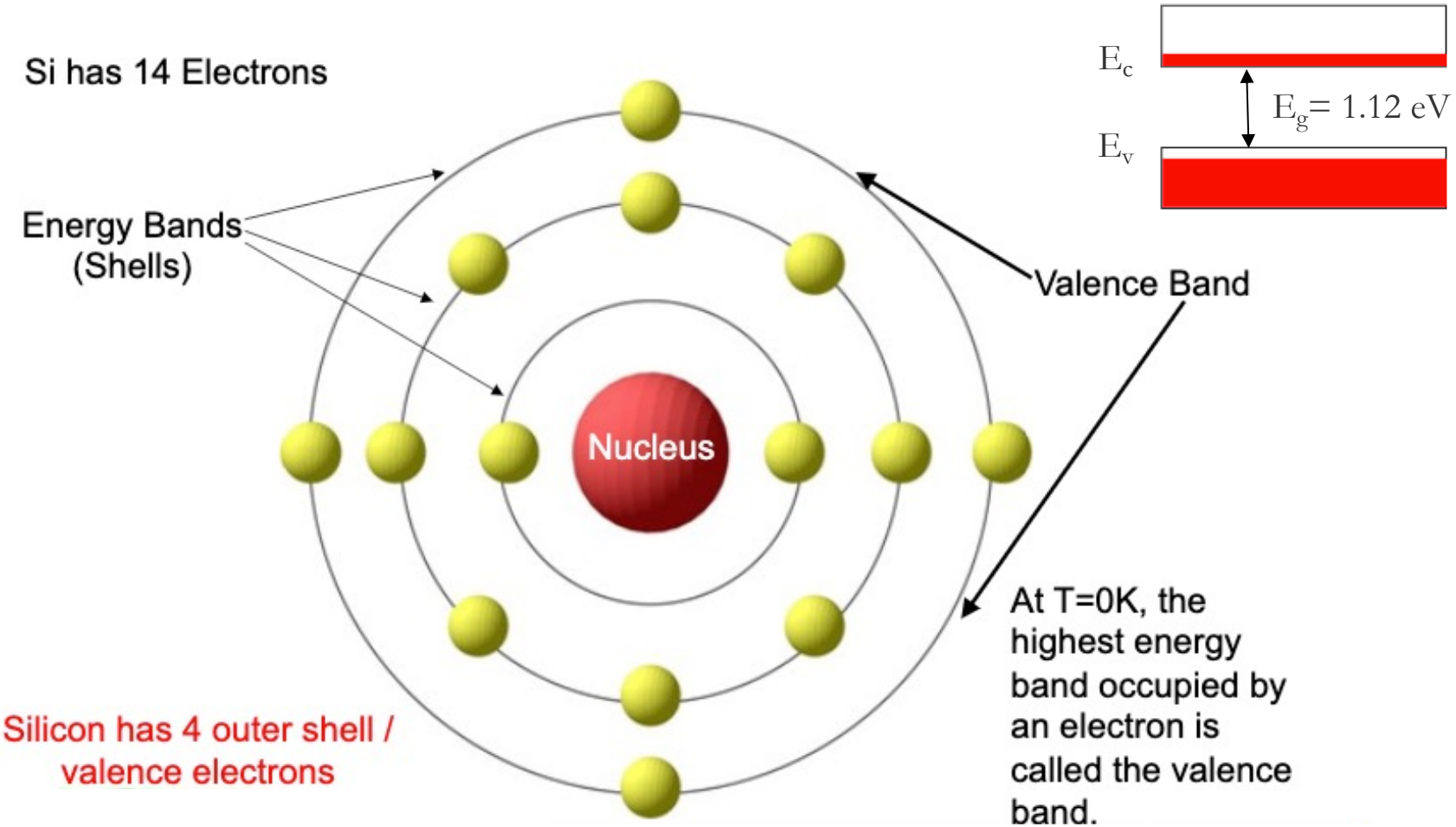
Semiconductor	Bandgap Energy E_G (eV)
Carbon (diamond)	5.47
Silicon	1.12
Germanium	0.66
Tin	0.082
Gallium arsenide	1.42
Gallium nitride	3.49
Indium phosphide	1.35
Boron nitride	7.50
Silicon carbide	3.26
Cadmium selenide	1.70

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

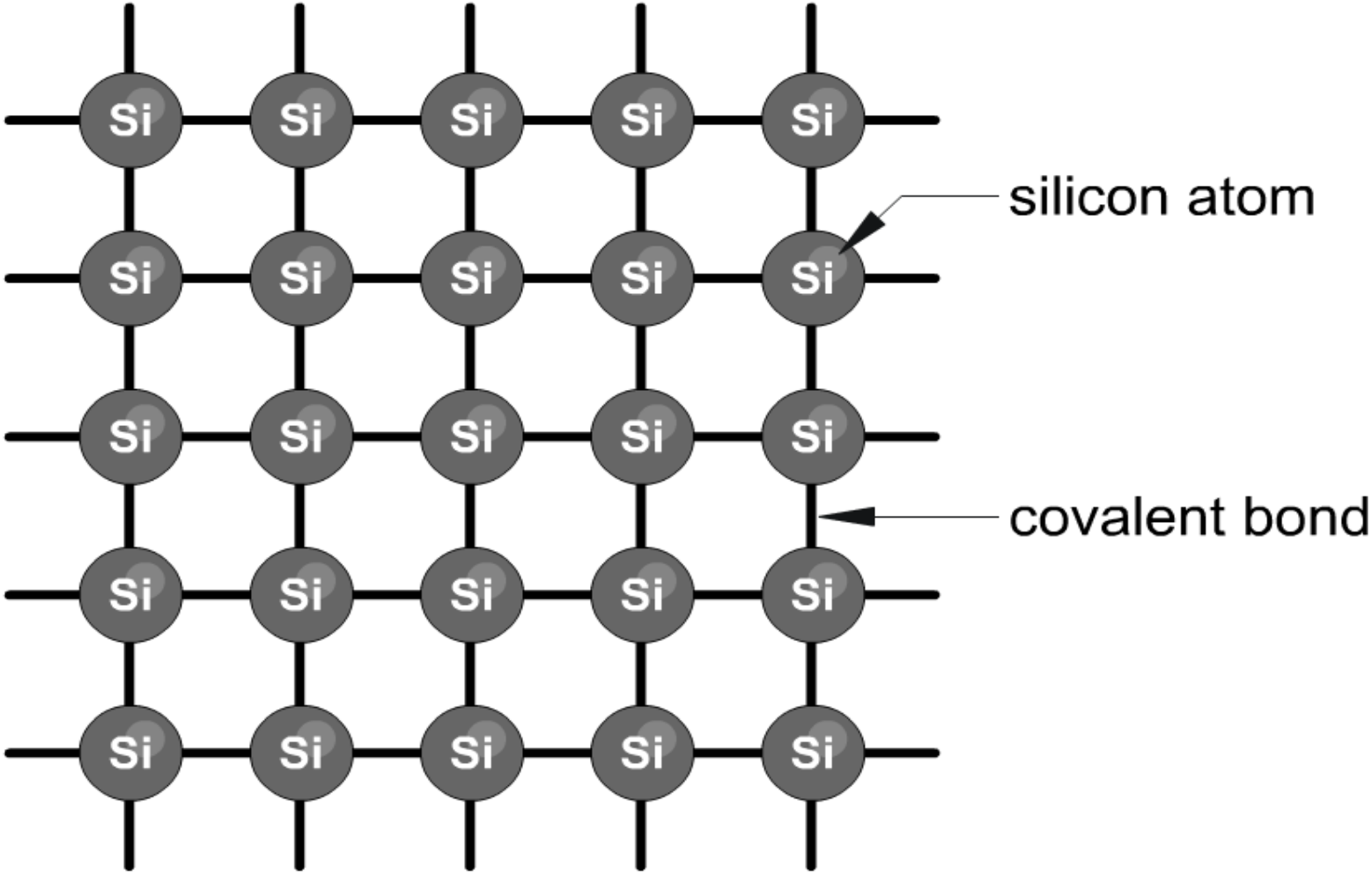
Example: Silicon



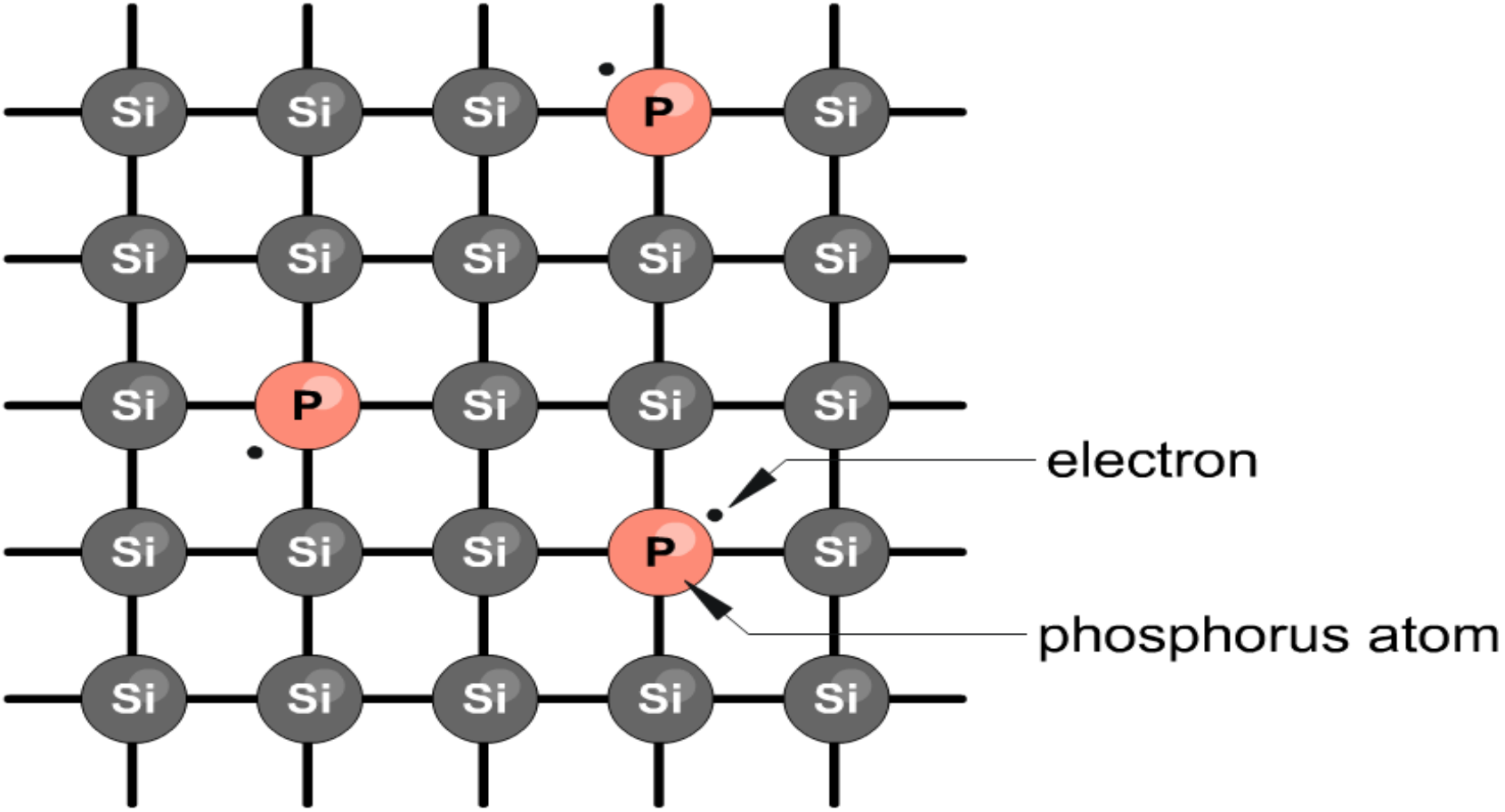
The Silicon Atom



Covalent Bonding in pure Si

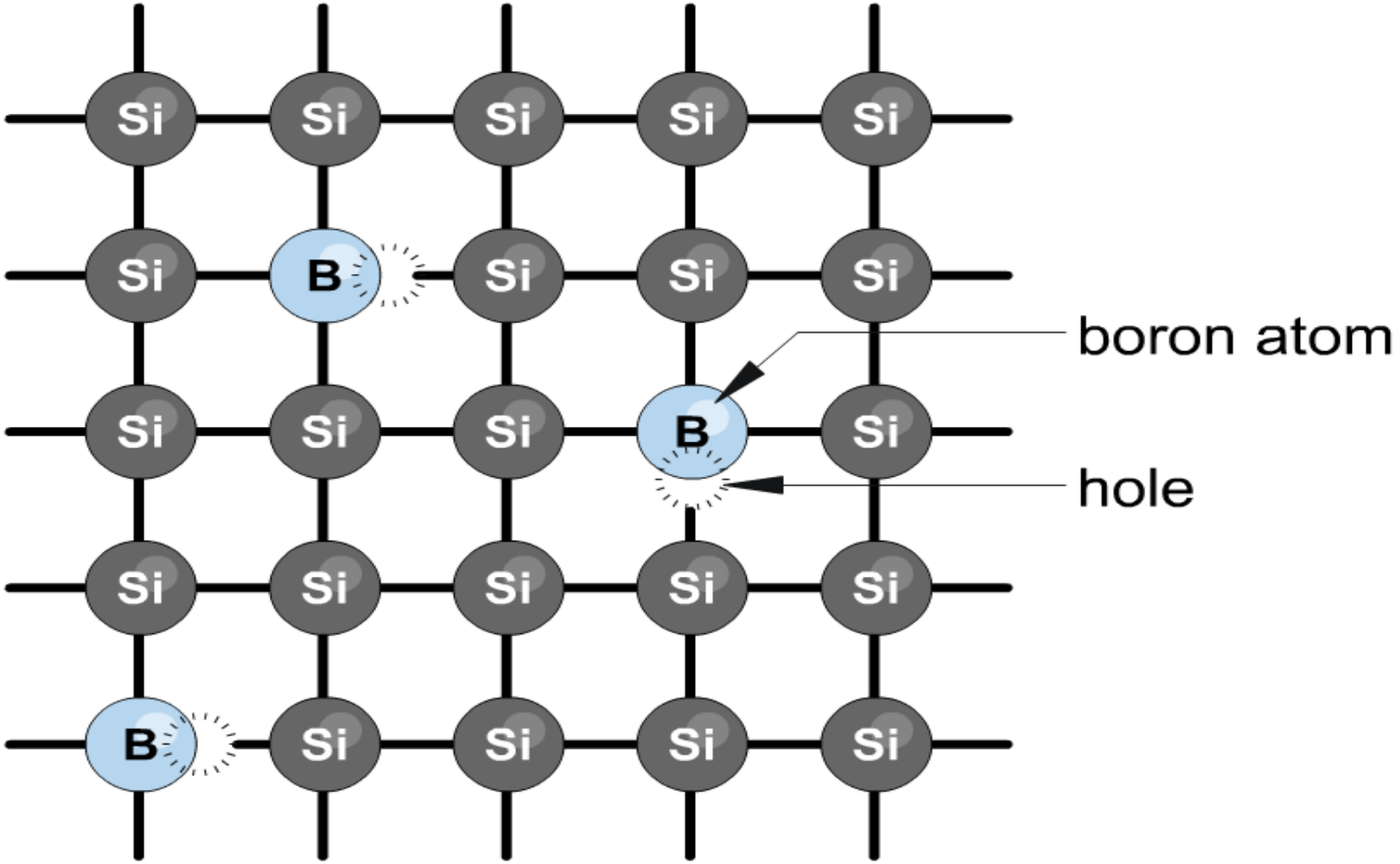


Doping: N-type Si



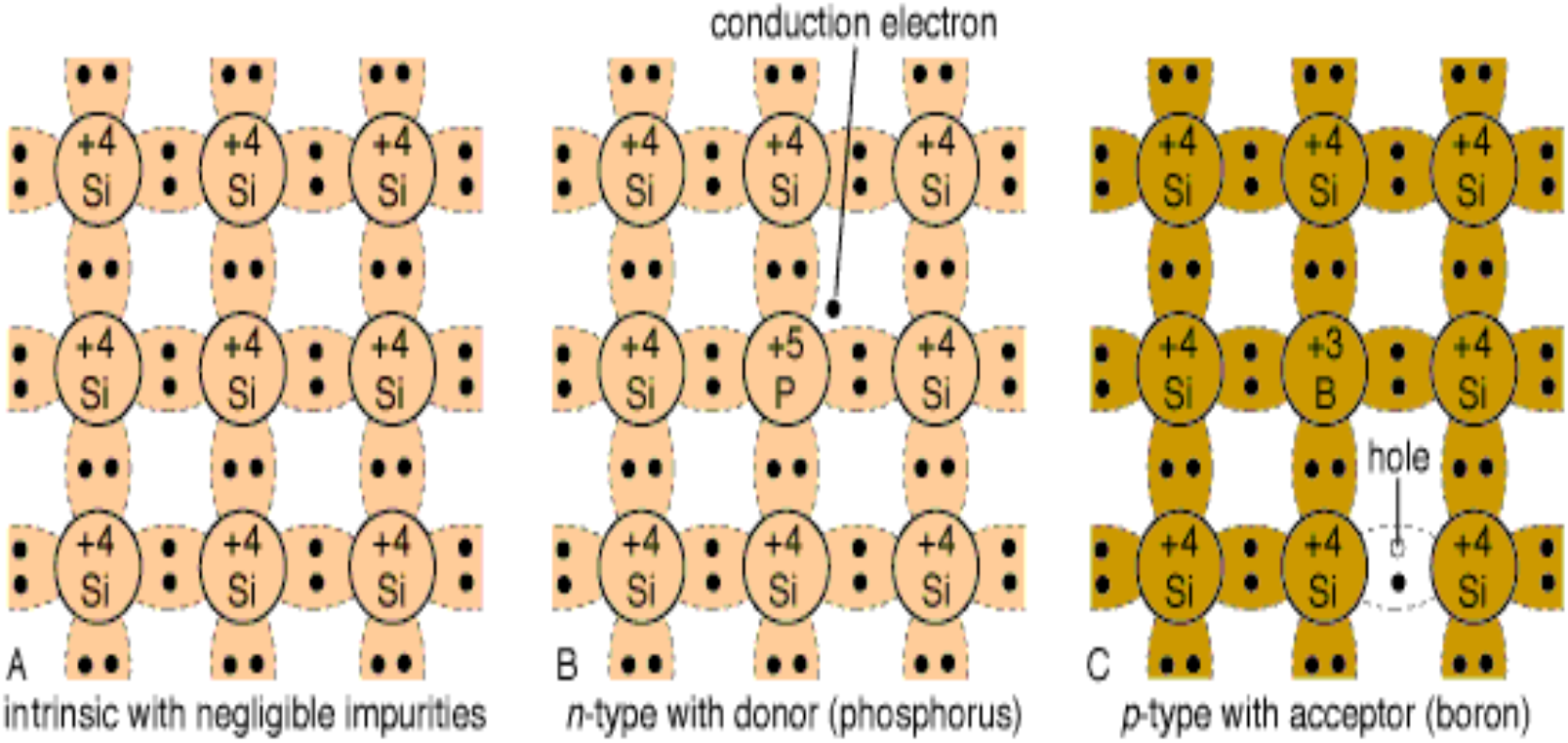
Donor concentration (N_D) 10^{15}-10^{18} atom/cm^3

Doping: P-type Si

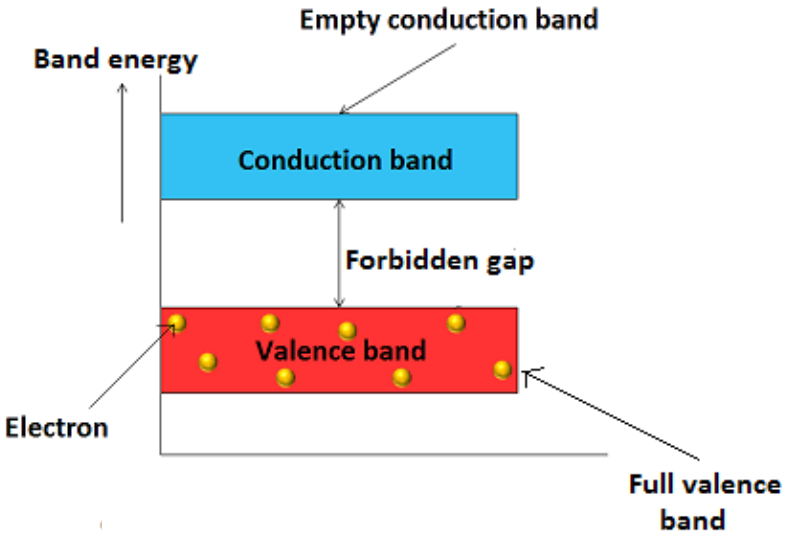


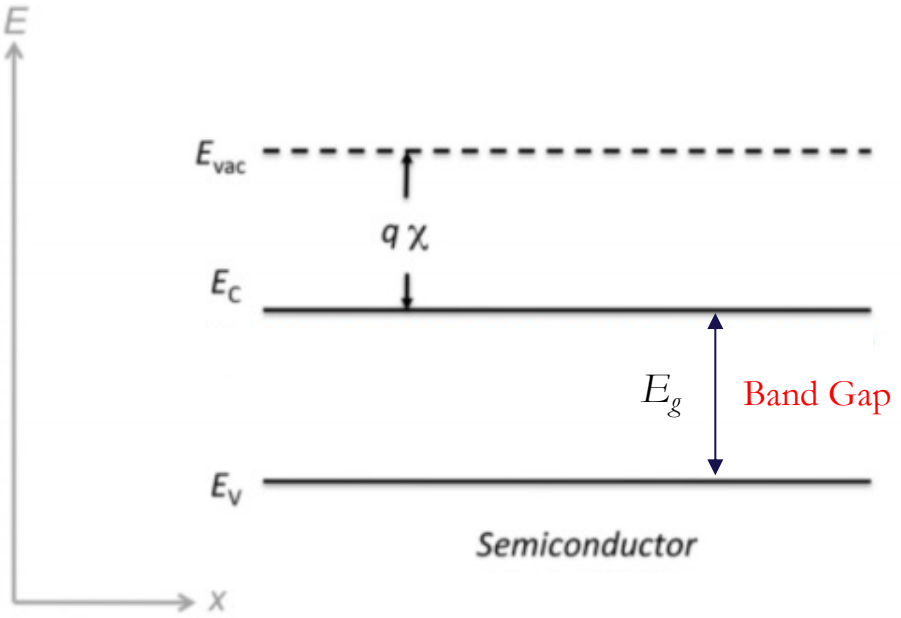
Acceptor concentration (N_A) 10^{13}-10^{16} atom/cm^3

Summary


conduction electron
+4 Si
+5 P
+3 B
hole
A
intrinsic with negligible impurities
B
n-type with donor (phosphorus)
C
p-type with acceptor (boron)

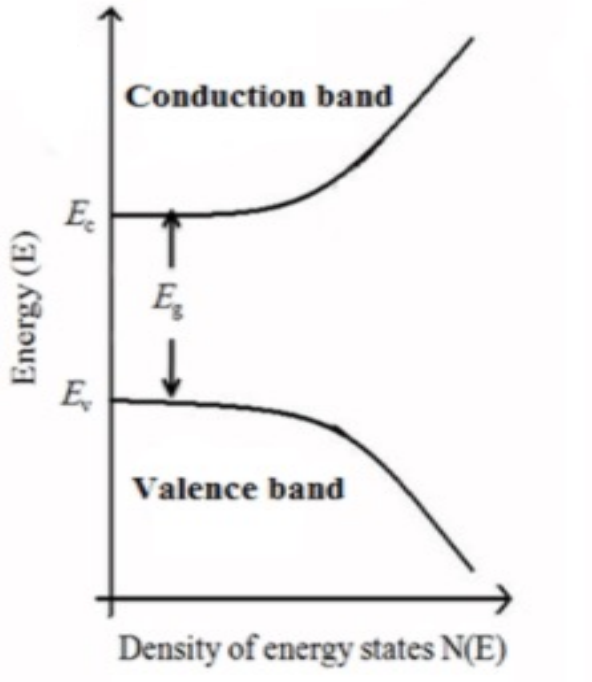
Back to the Energy Bands

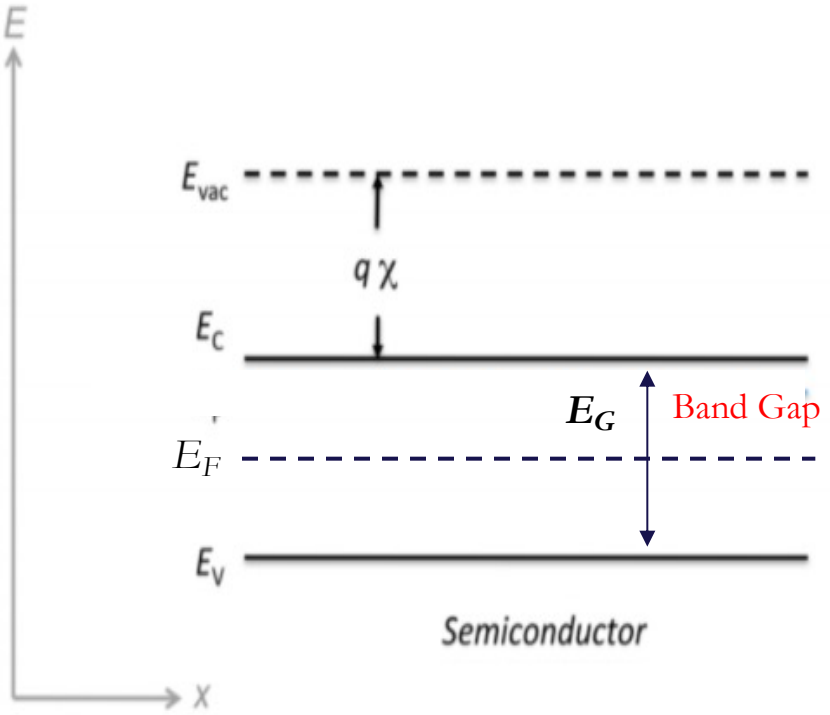




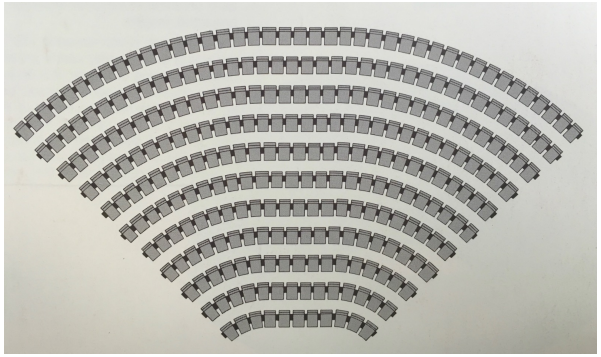
Band Diagram

Back to the Energy Bands



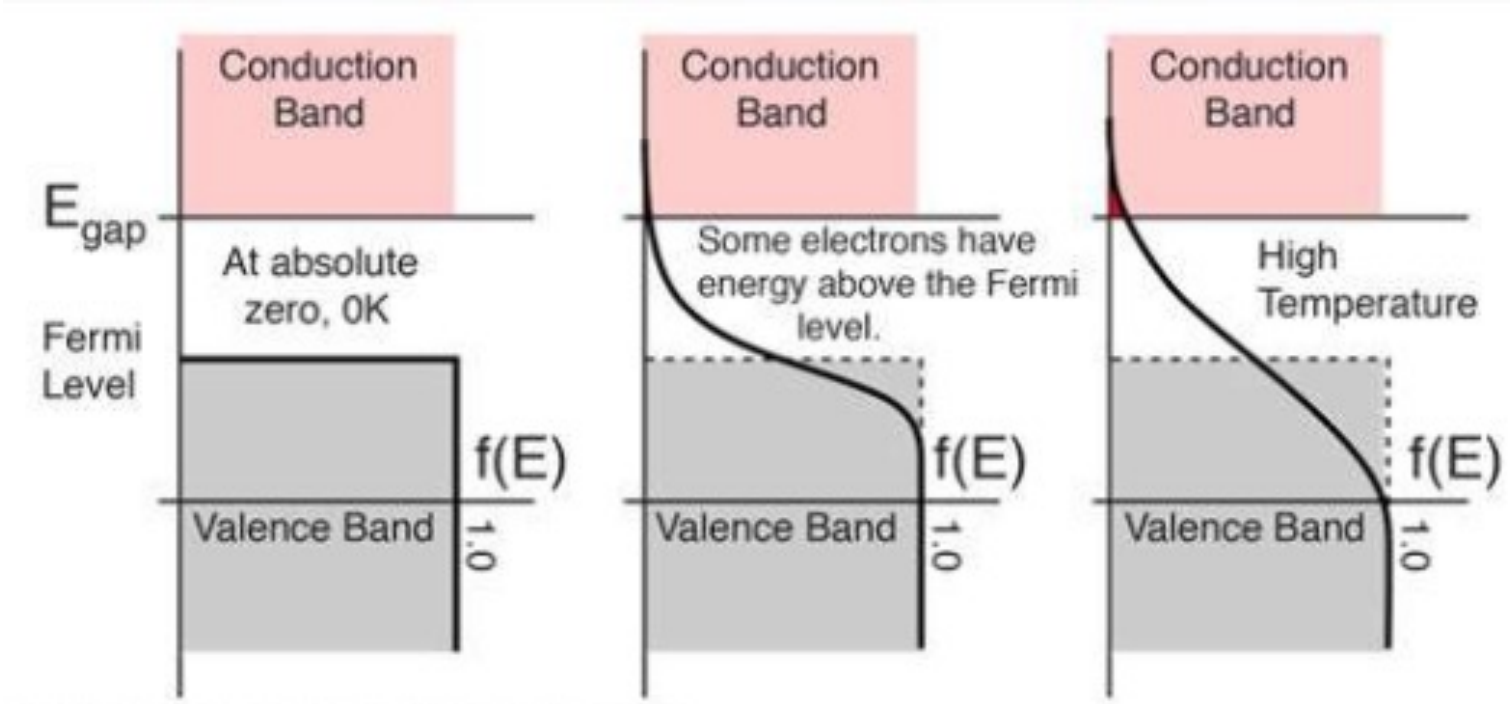


Density of States

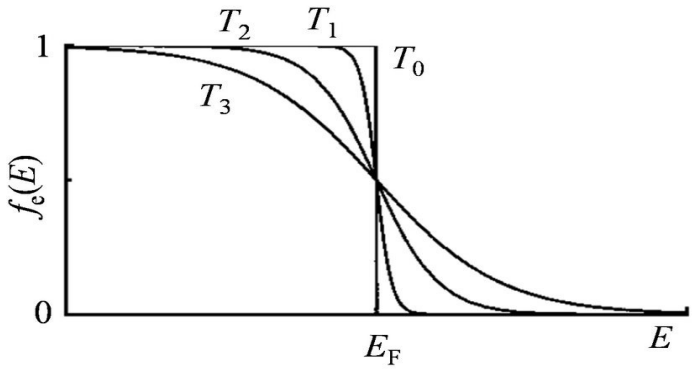


Band Diagram

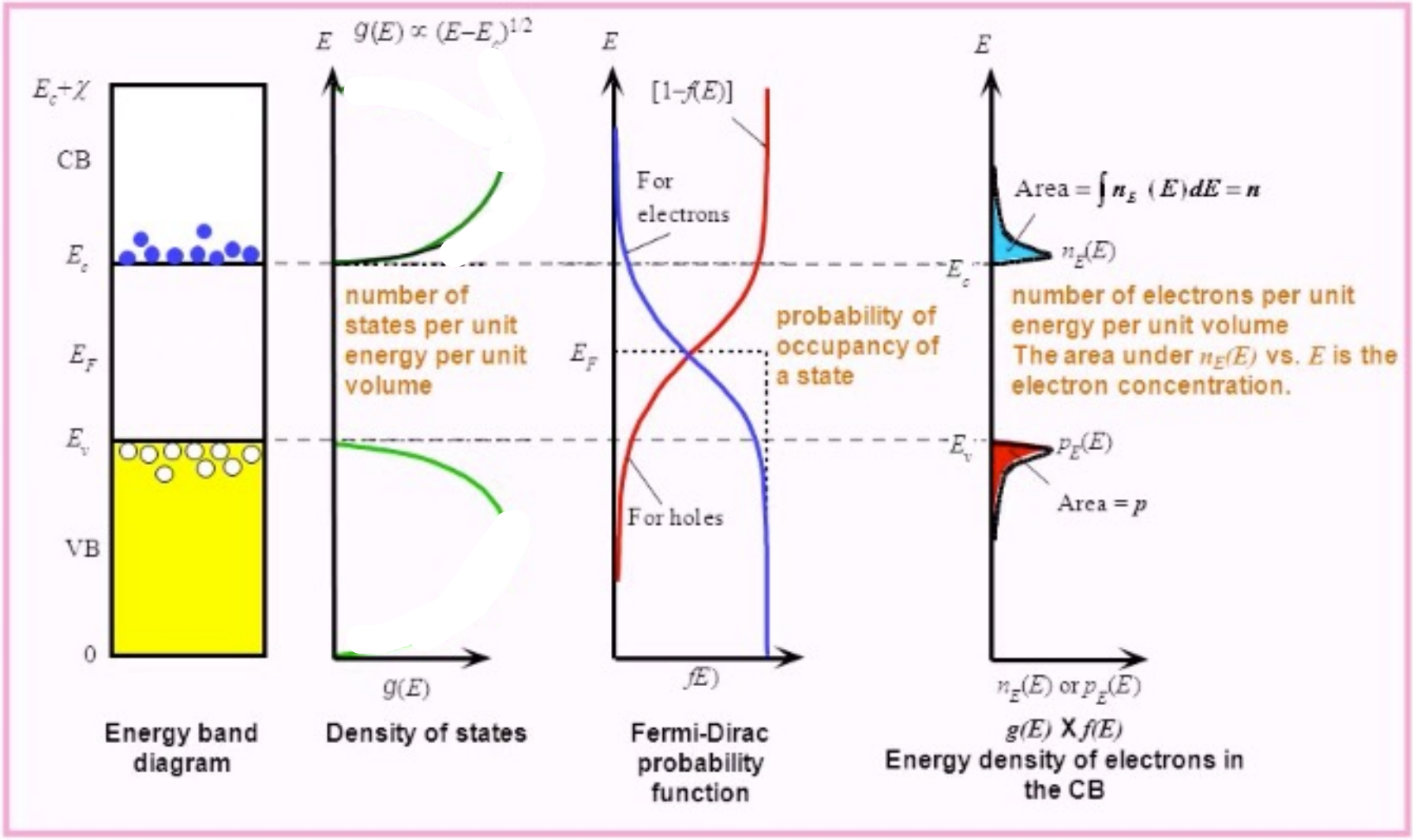
Fermi Distribution



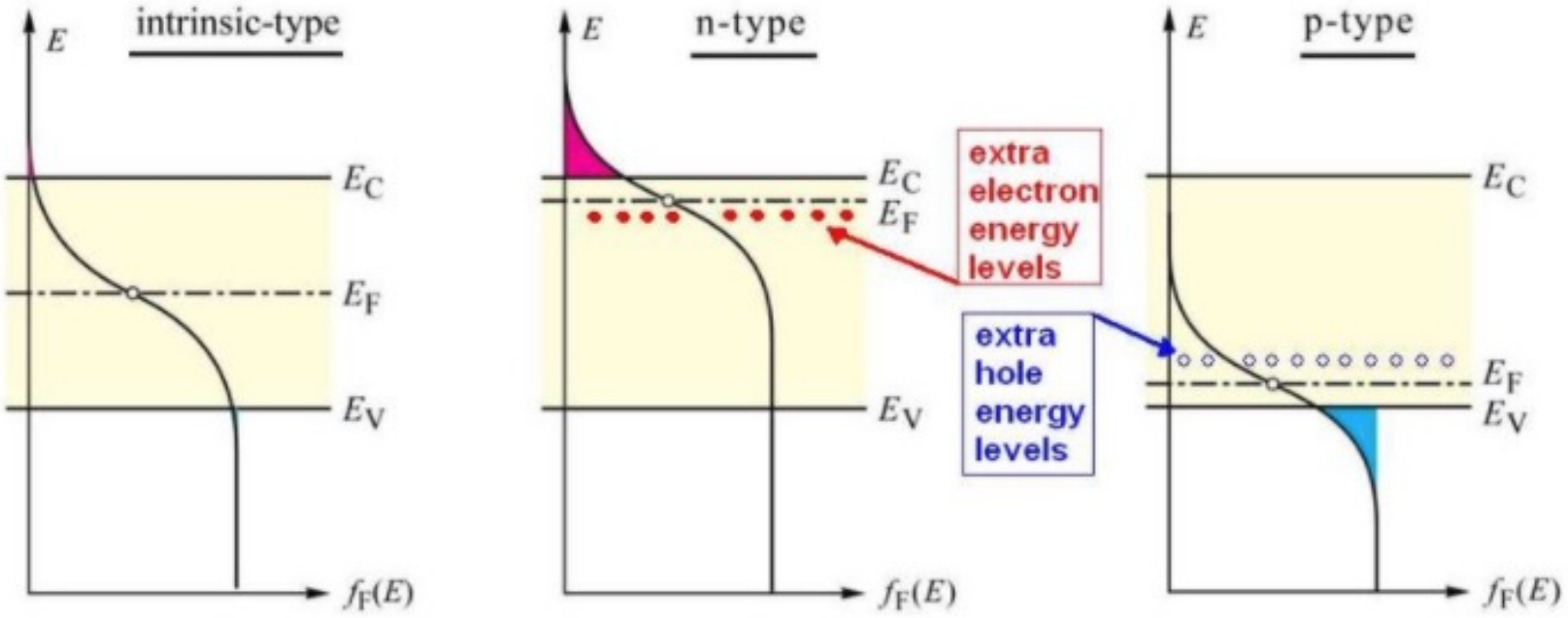
$$f_F(E) = \frac{1}{1 + exp\left(\frac{E - E_F}{kT}\right)}$$



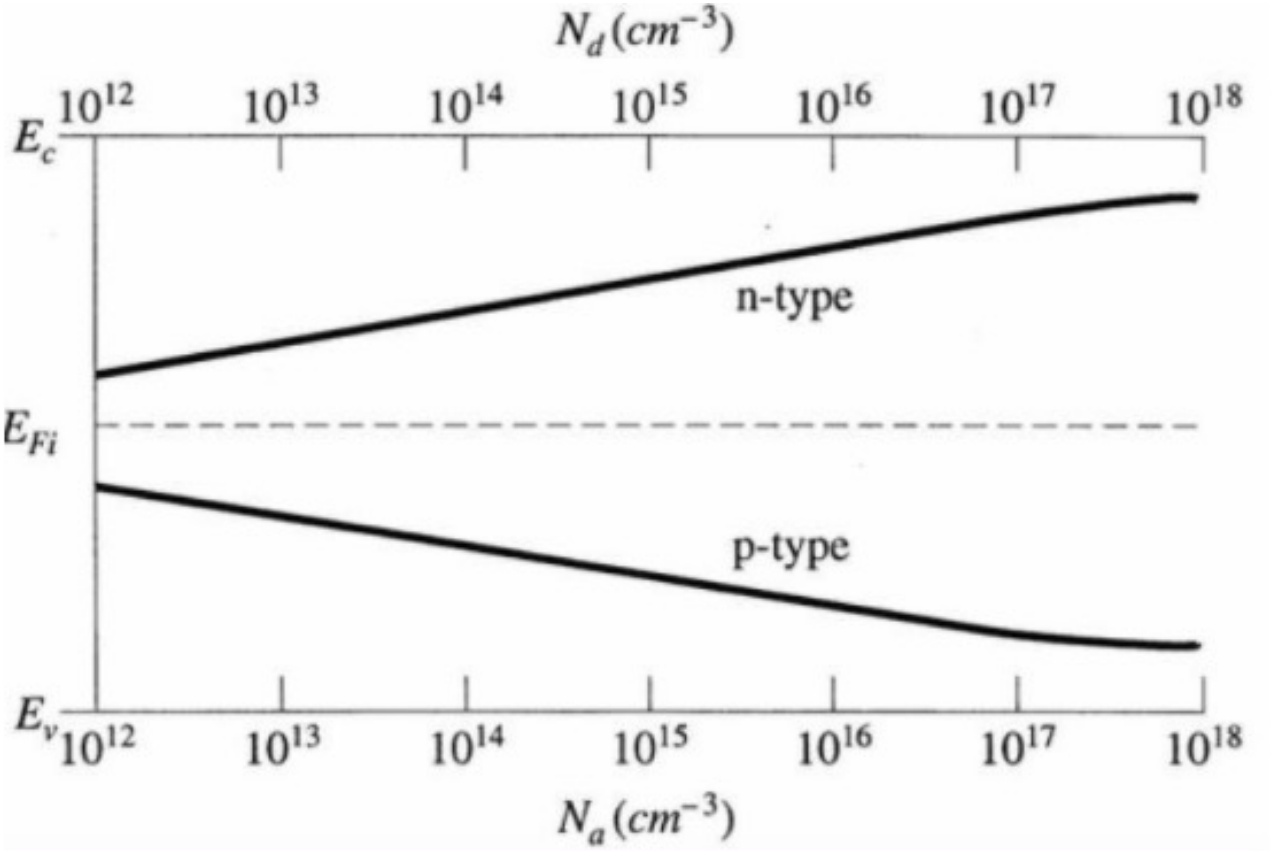
Electron/Hole Density



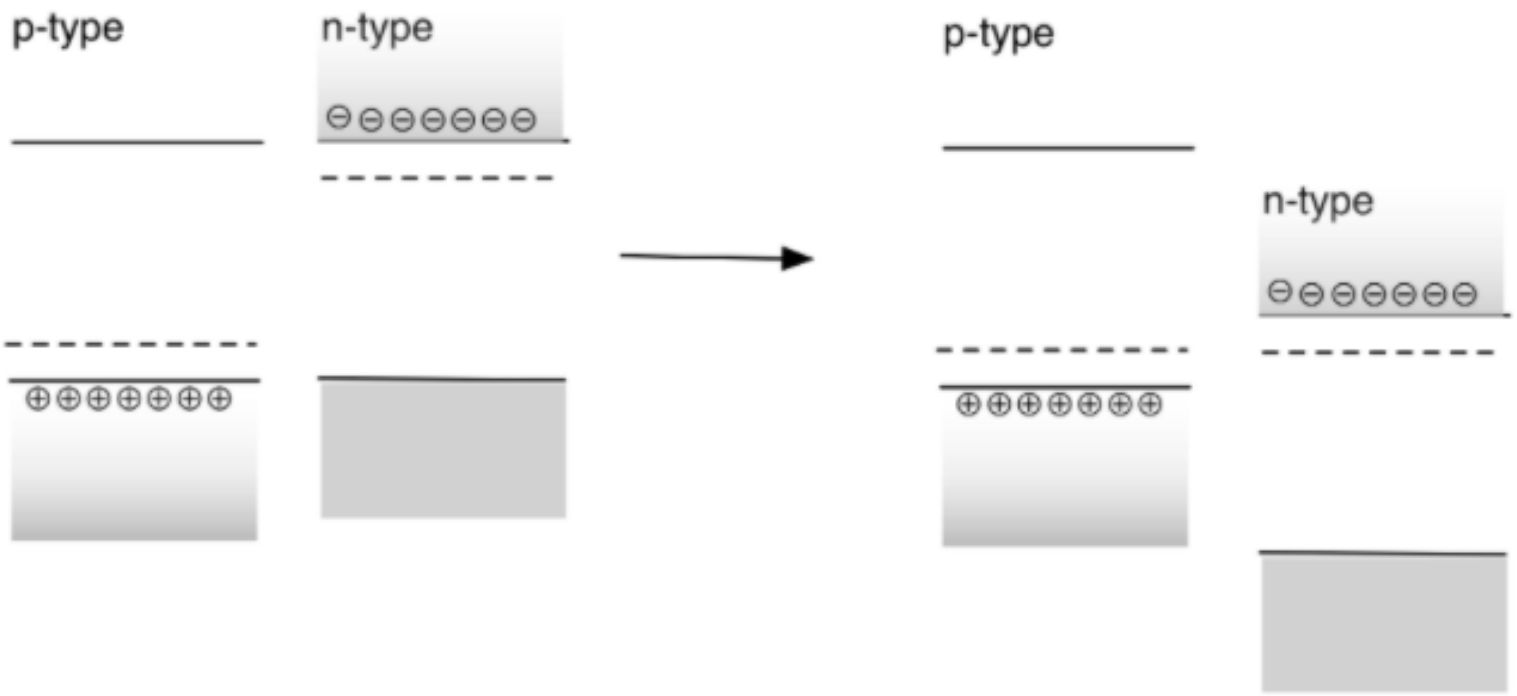
The n-type and p-type Band Diagram



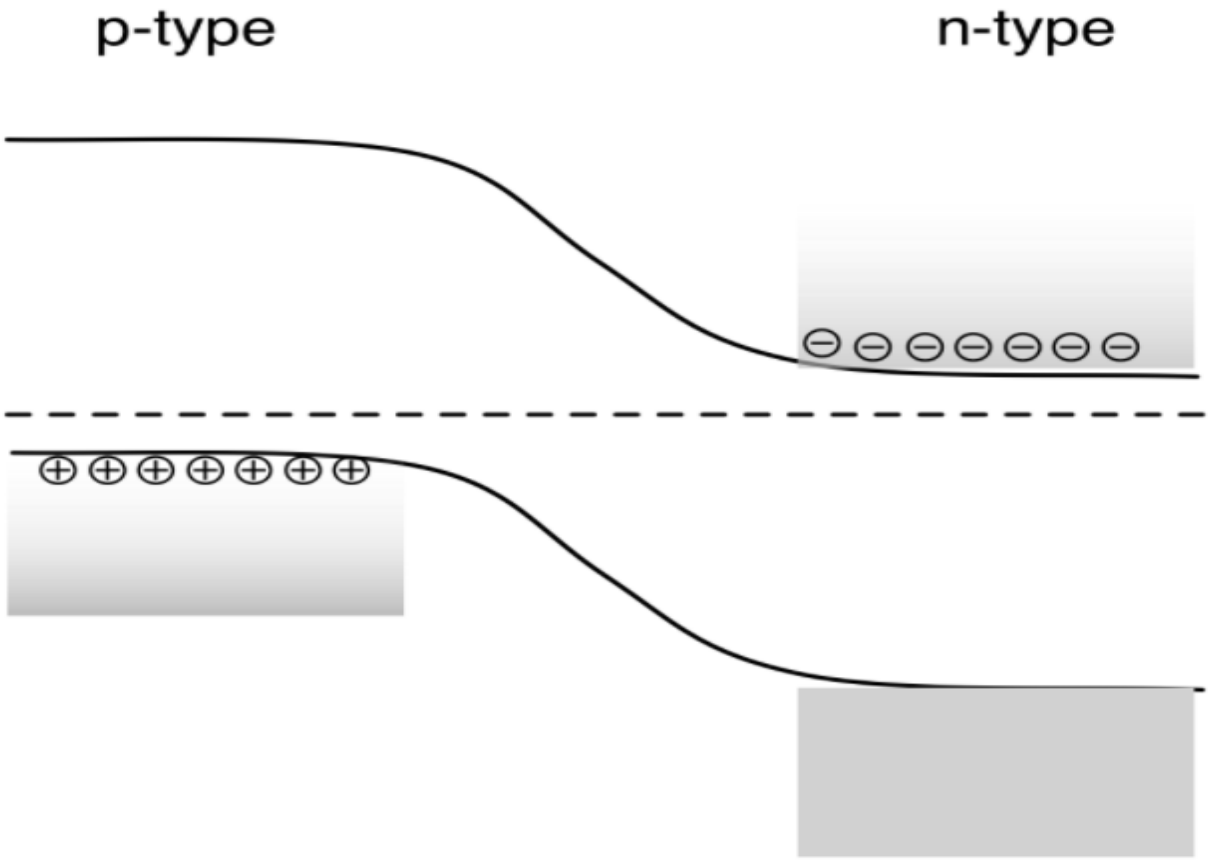
The n-type and p-type Band Diagram



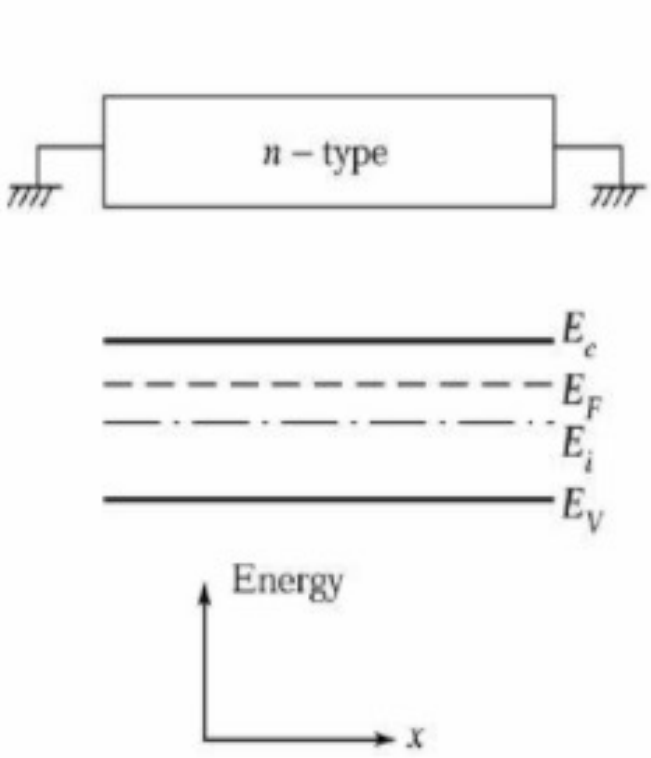
The PN Junction Band Diagram

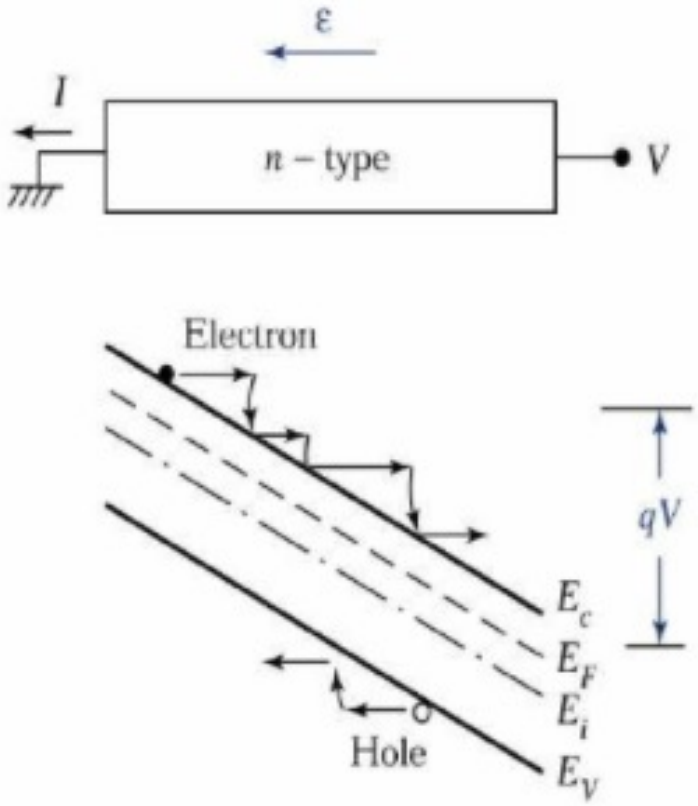


Unbiased PN Junction Band Diagram

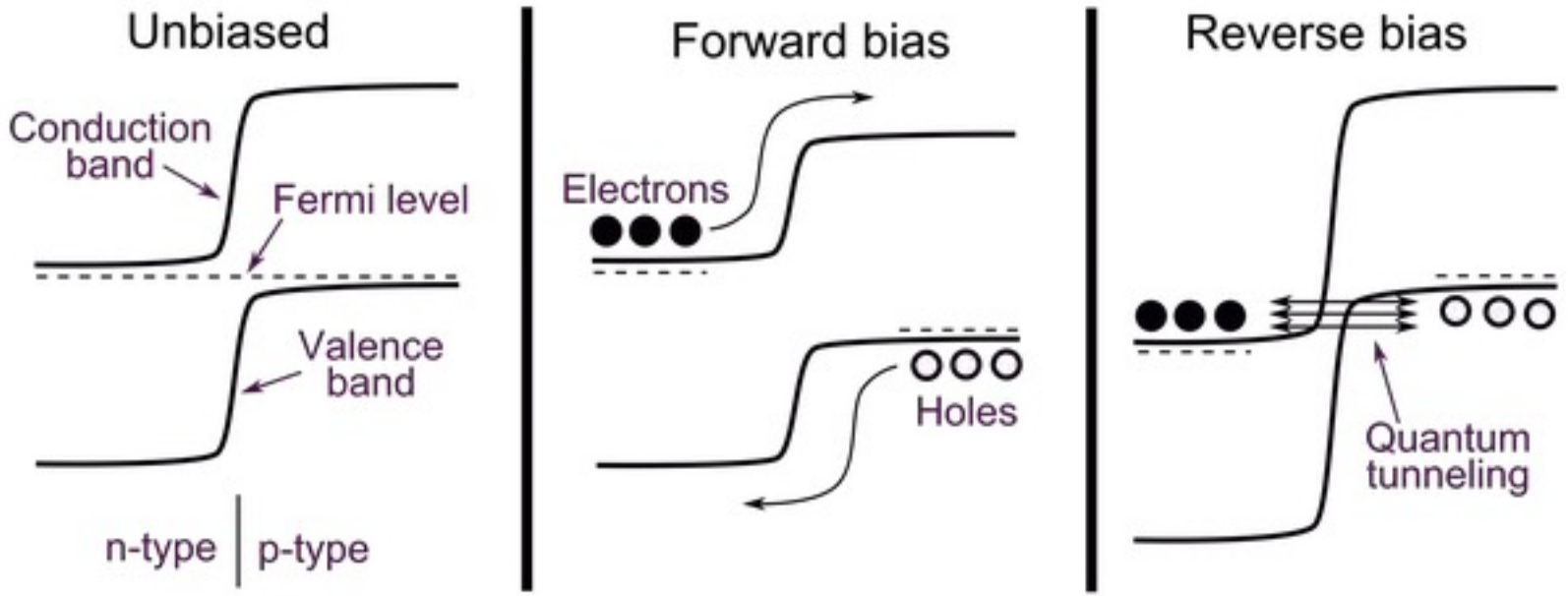


Biased n-type Band Diagram



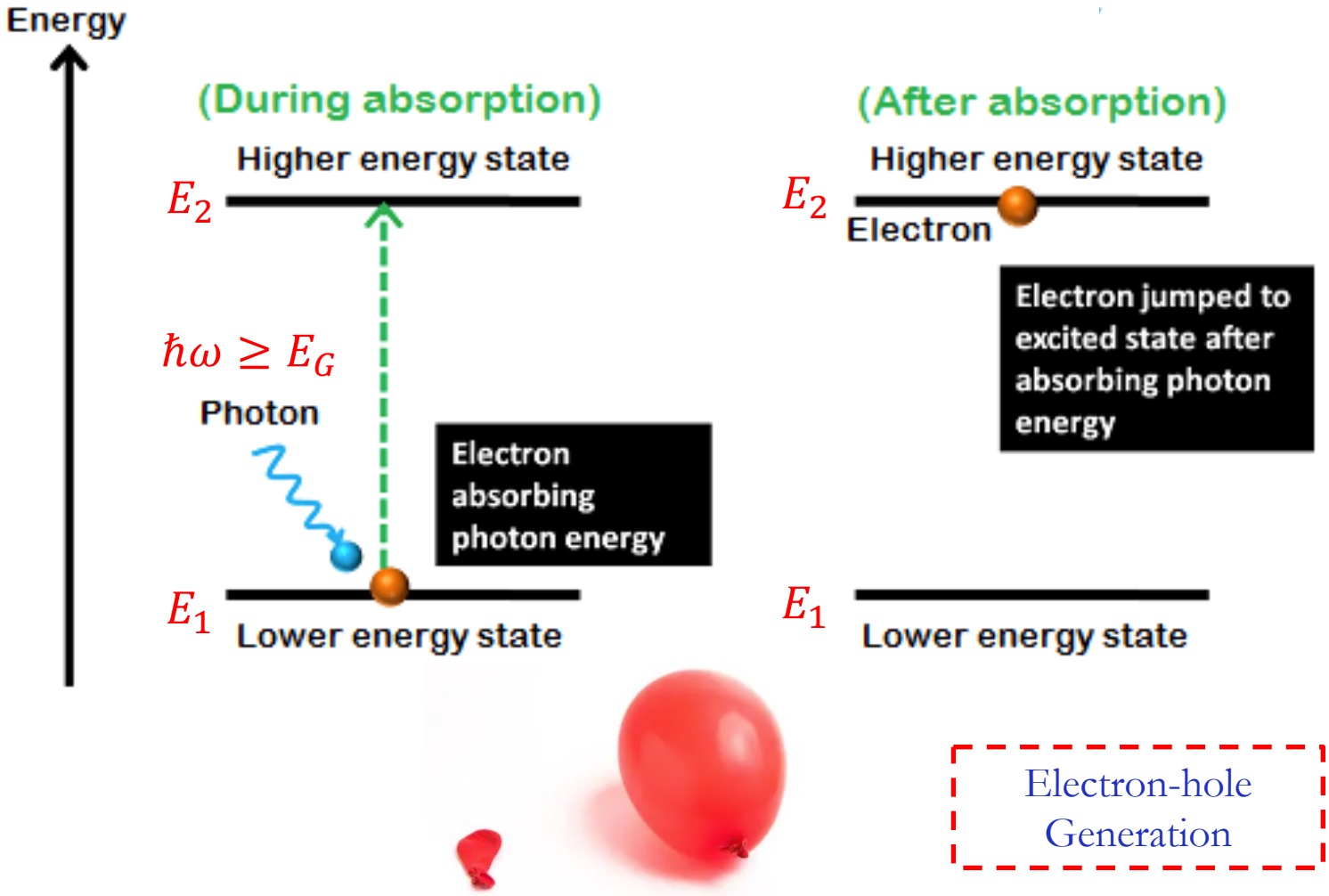


Biased PN Junction Band Diagram



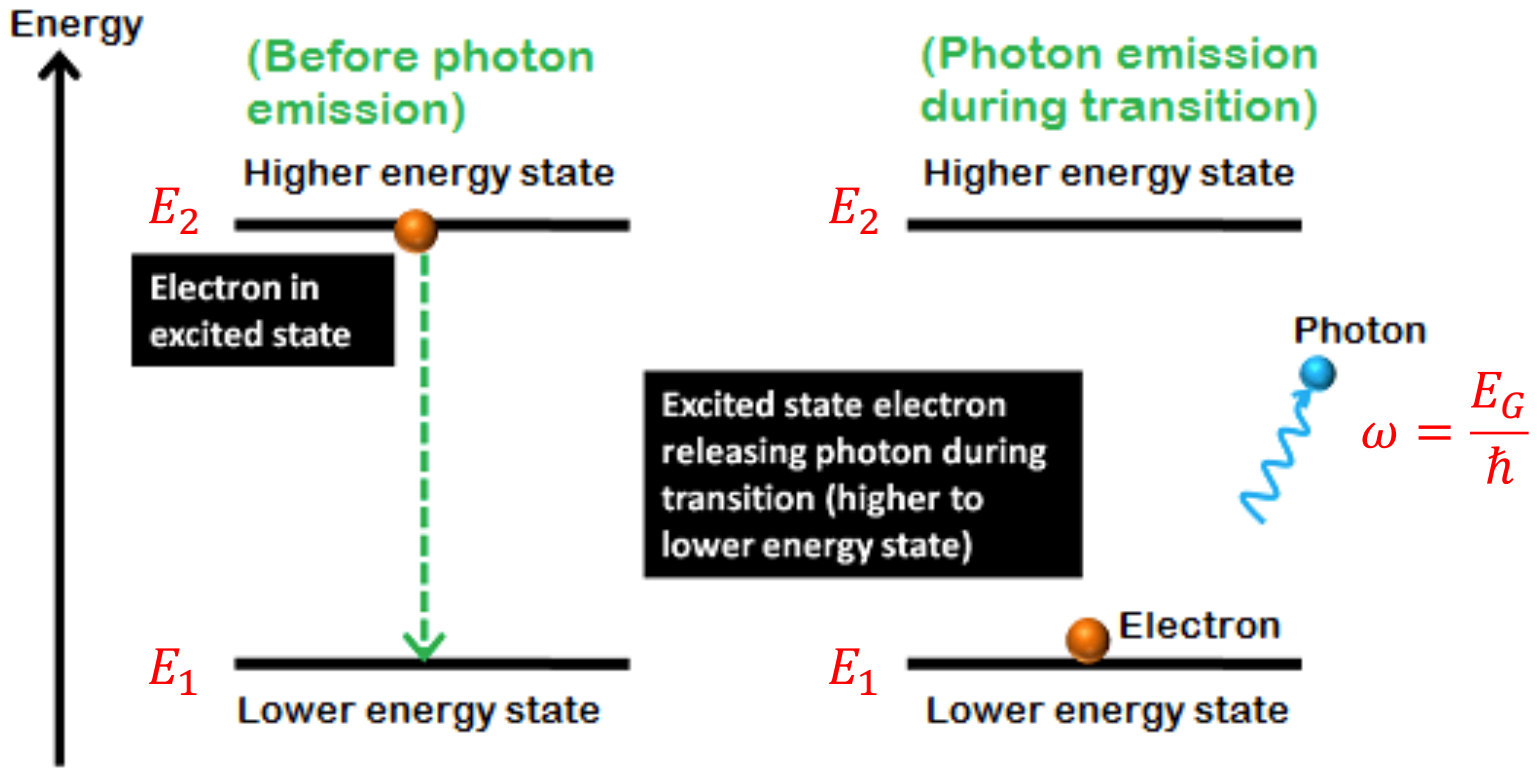
Photon Generation and Absorption

Absorption



Photon Generation and Absorption

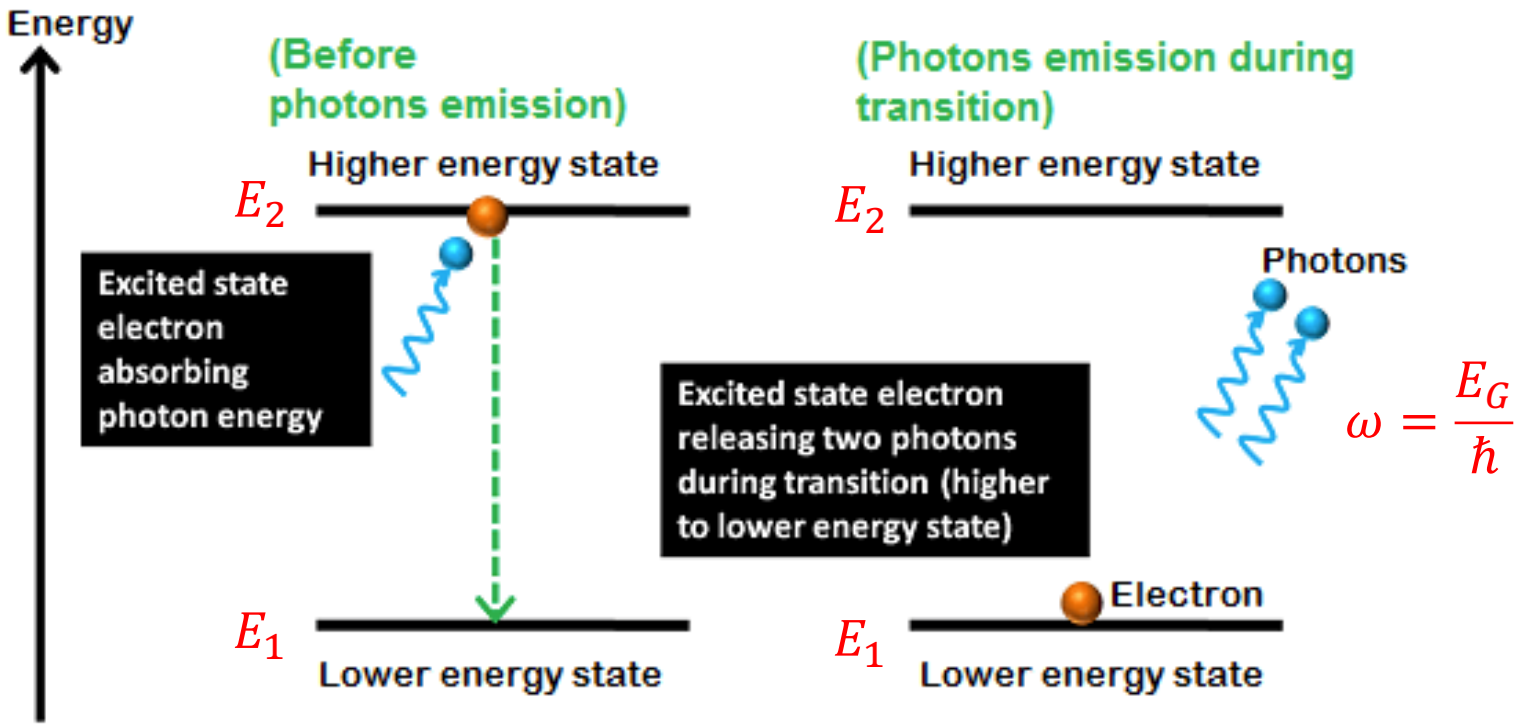
Spontaneous Emission



Electron-hole Re-combination (radiative/nonradiative)

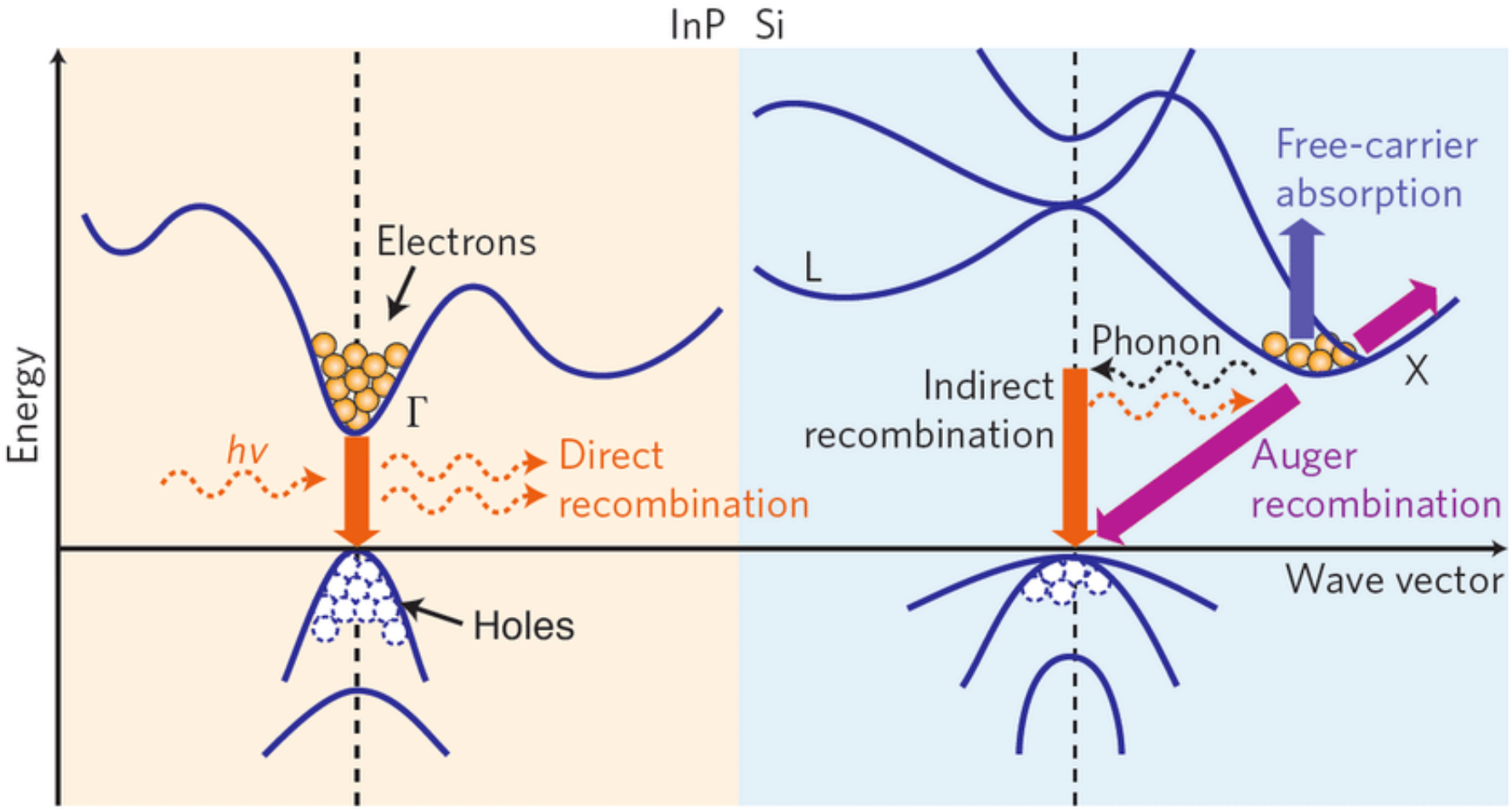
Photon Generation and Absorption

Stimulated Emission



Electron-hole
Re-combination
(radiative/nonradiative)

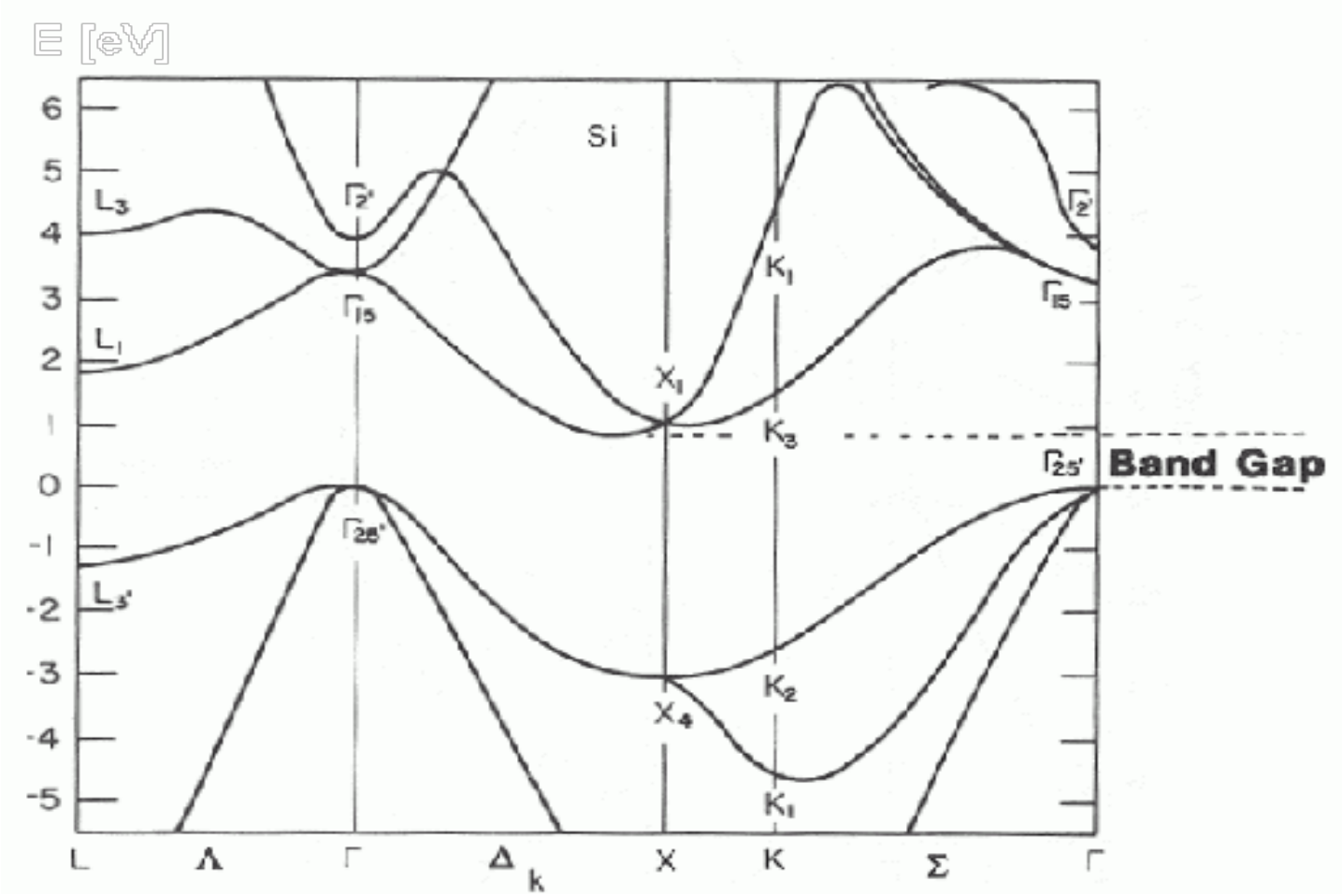
Direct and Indirect Gap Materials



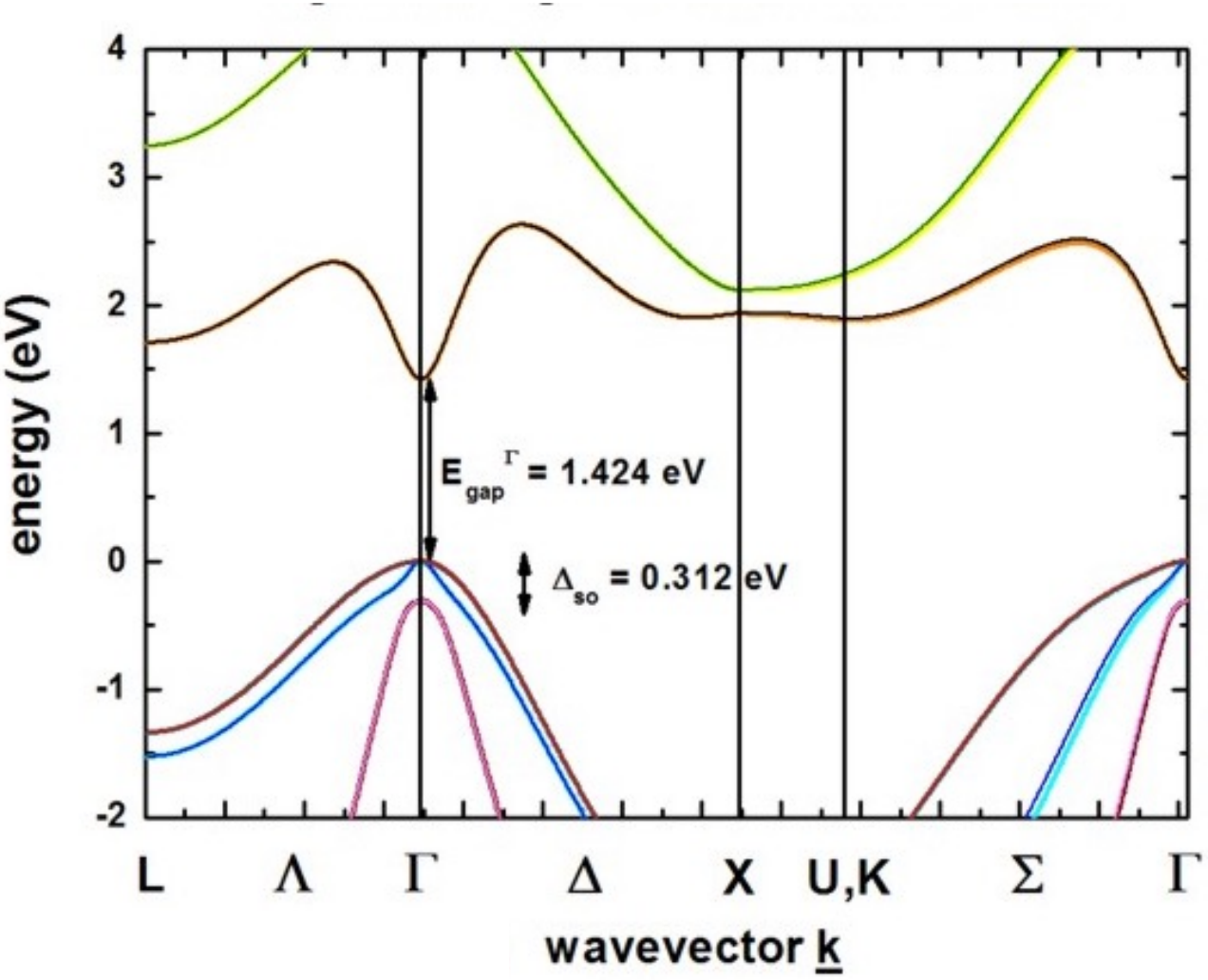
Direct and Indirect Gap Materials

	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

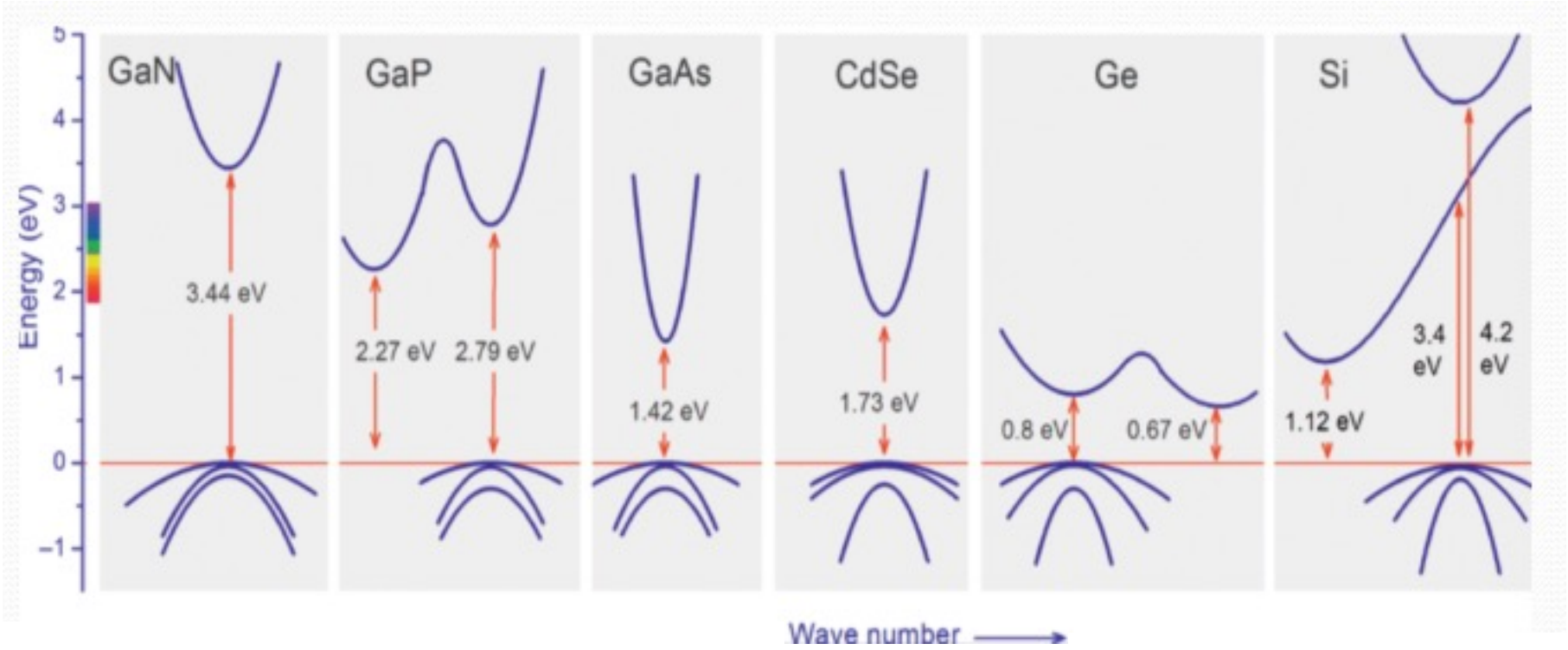
Si: Indirect Gap Material



GaAs: Direct Gap Material



GaAs: Direct Gap Material



Bandgap Engineering

Example: Emission in GaAs, GaN and InN.

$$\omega = \frac{E_G}{\hbar}$$

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega}$$

$$\lambda = \frac{2\pi c\hbar}{E_G}$$

$$\lambda_g = \frac{hc}{E_G}$$

GaAs band gap energy E_G=1.42 eV

$$\lambda_g = \frac{2\pi(3\times10^8)(1.055\times10^{-34})}{(1.42)(1.6\times10^{-19})}$$

$\lambda_g = 875\ nm$ NIR

GaN band gap energy E_G=3.49 eV

$\lambda_g = 356\ nm$ UV

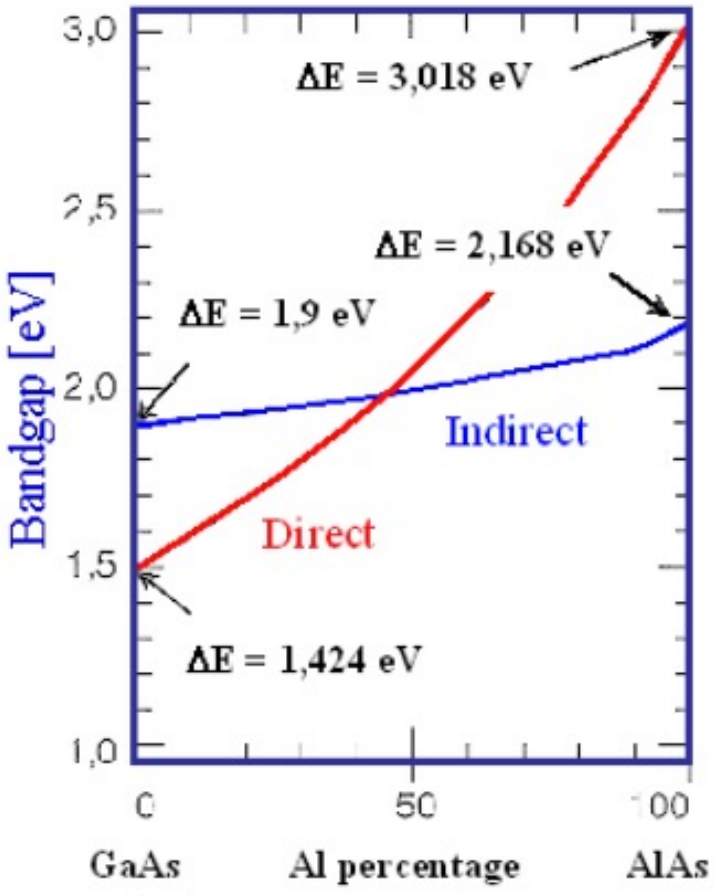
InN band gap energy E_G=0.69 eV

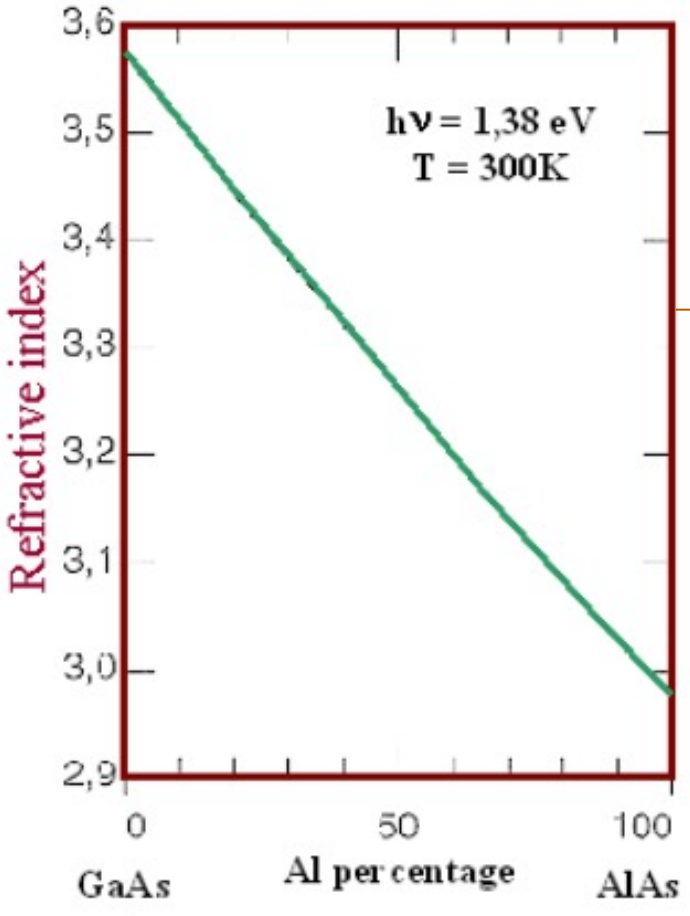
$\lambda_g = 1800\ nm$ IR

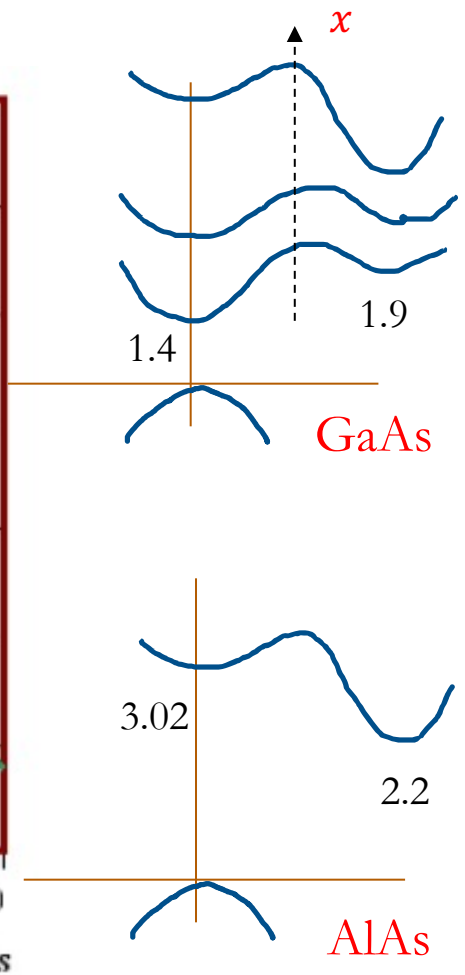
The alloy $In_xGa_{1-x}N$ is very useful for spanning the visible range.

Bandgap Engineering

Example: Emission in $Al_xGa_{1-x}As$.





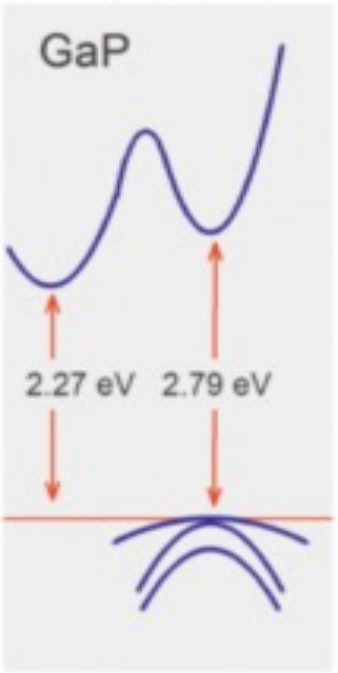


The Green Gap

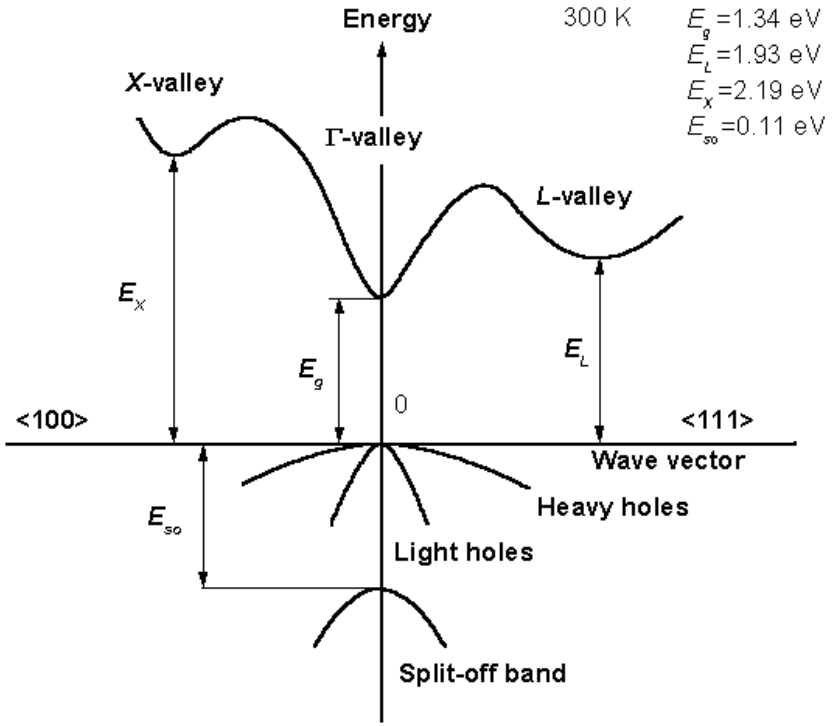
Green Band: 530-570 nm.

Green Band: 2.2-2.3 eV.

Emission in $Ga_xIn_{1-x}P$.



InP



Photodetectors

The following subjects will be covered

- Photodetector Types.
- General Characteristics of Photodiodes.
- Modes of Operation.
- PN, PIN and APD Devices.

Photodetectors

Type (1): Photo-emissive

Light is absorbed, electron-hole pair generated and ejected out of material as current.

Type (2): Photo-conductive

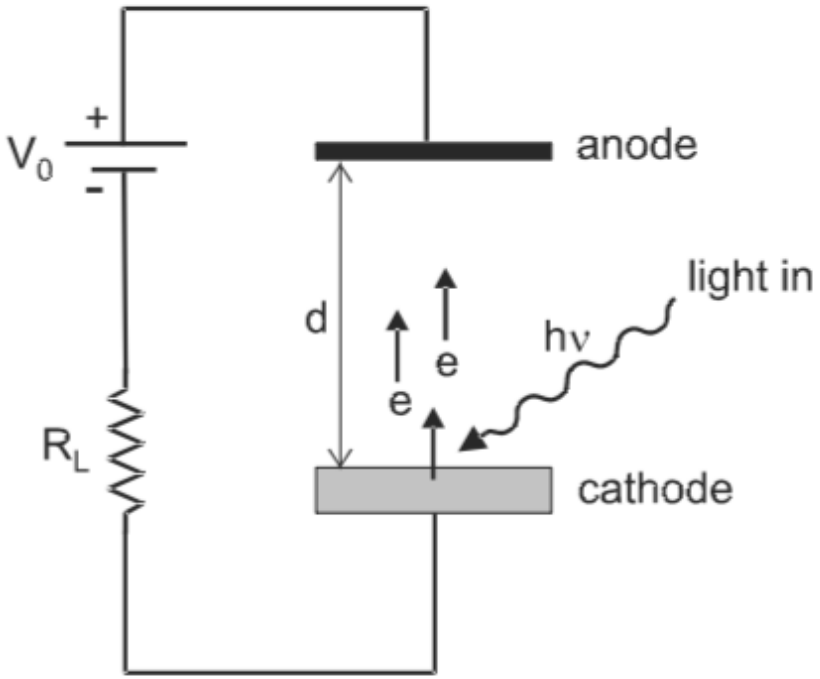
Light is absorbed, electron-hole pair generated and the material conductivity is changed.

Examples:

- Photoconductors: A piece of semiconductor the conductance of which is controlled by light (photoresistors).
- Photodiodes: A pn junction-based device that converts light energy to electron-hole pairs. The ehp are saved and used for operation. (PIN diodes, APDs, Solar cells, etc.)

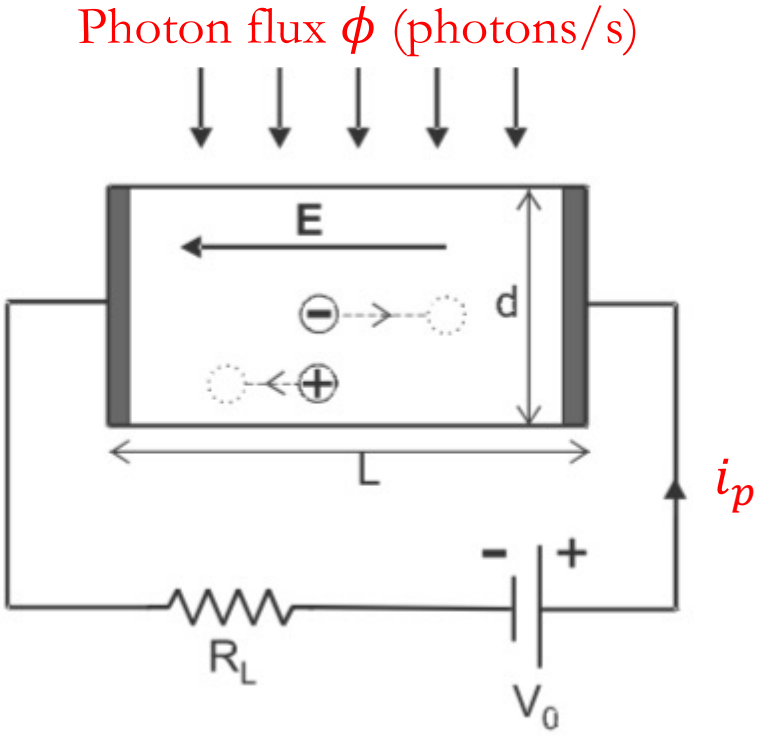
Photo-emissive Devices

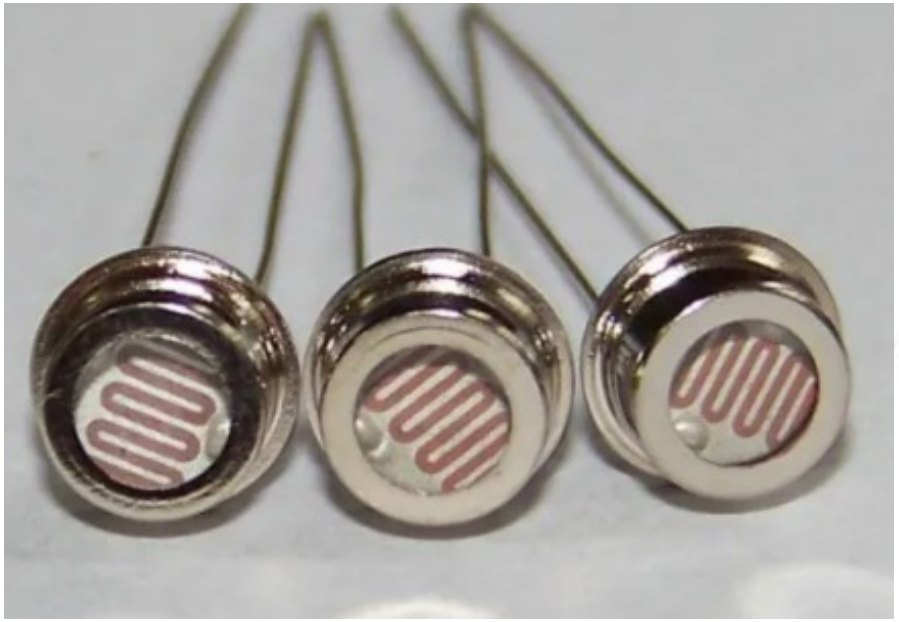
The basic structure of a photo-emissive device is a photoelectric-effect set-up.



Photoconductive Devices

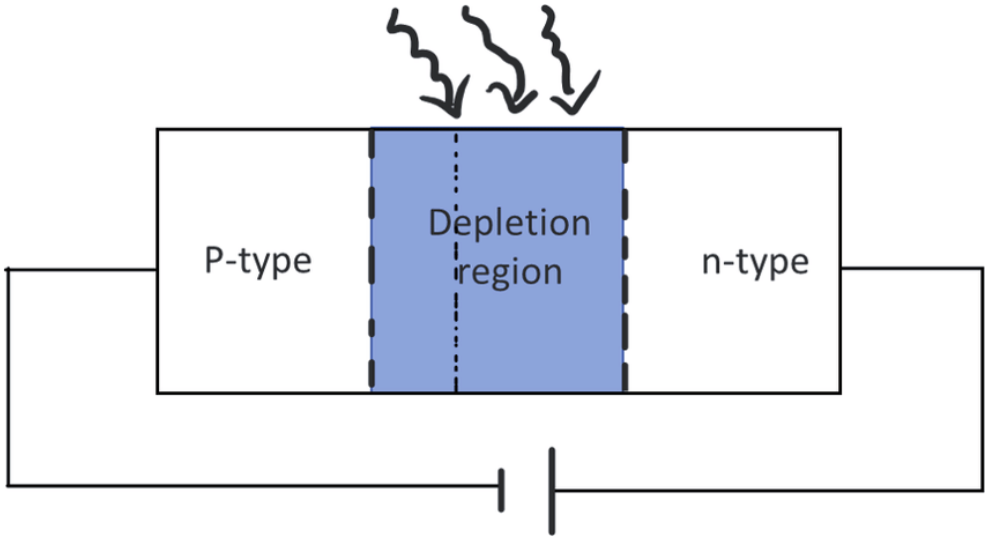
The basic structure of a photoconductive device is a semiconductor material connected across a battery.





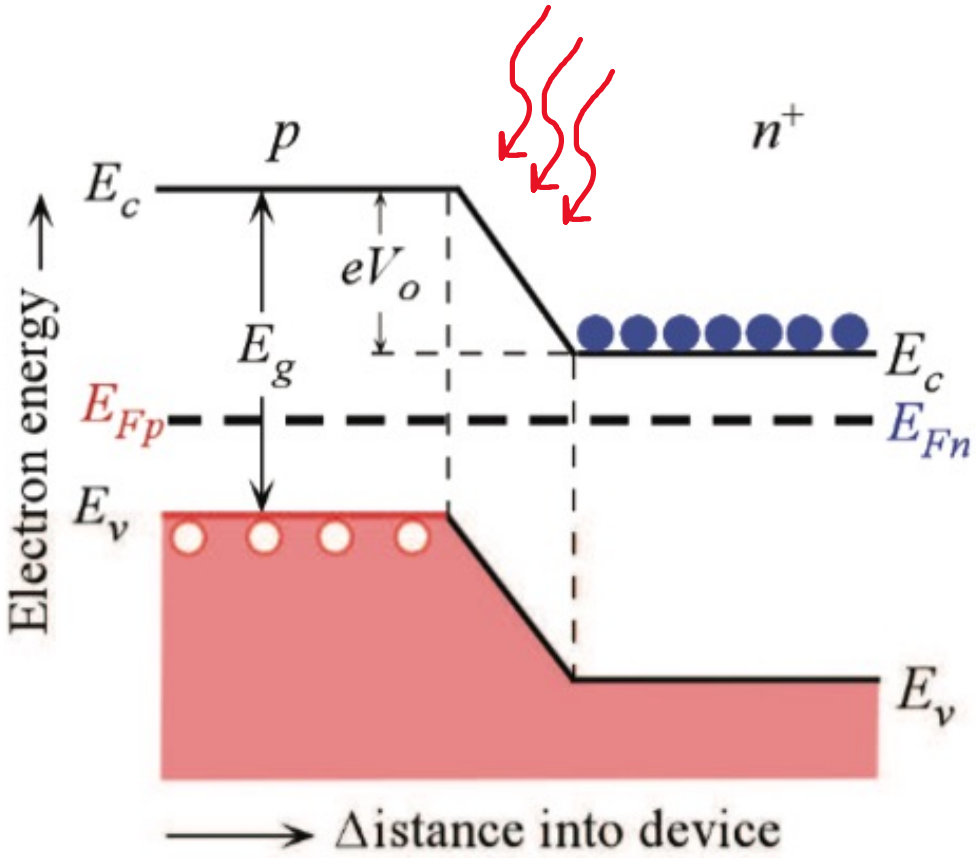
Photodiodes

The basic structure of a photodiode device is a pn junction.



Photodiodes

The basic structure of a photodiode device is a pn junction.



Photodiode Characteristics

Quantum efficiency. The ratio of optical current ehps to incident photon flux.

Responsivity. The ratio of optical current to incident optical power.

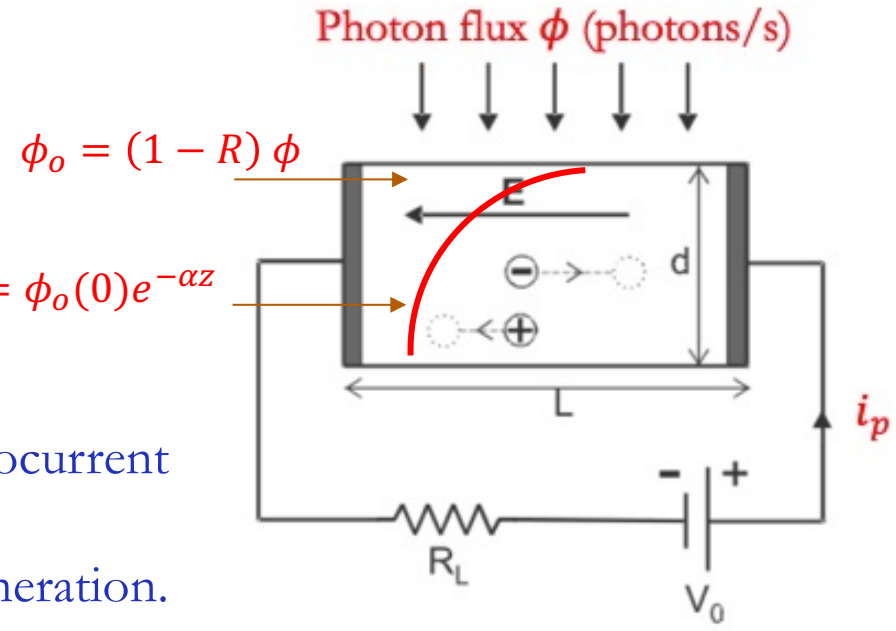
Impulse response. The rise time of the output current or the output bandwidth.

Noise power. The amount of current created by sources other than light.

Photodiode Characteristics: Efficiency

Quantum efficiency. The ratio of created ehp to number of incident photons.

$$\eta = \frac{\text{ehp flux making the optical current}}{\text{Incident photon flux } (\phi)}$$



Absorbed photon flux:

$$(1-R)\left(1-e^{-\alpha d}\right)\phi$$

Not all absorbed photons will make photocurrent due to:
(1) ehp recombination (2) phonon generation.
Introduce the factor $0 \le \xi \le 1$.

Current-making photon flux = $(1-R)\left(1-e^{-\alpha d}\right)\xi\,\phi$

$$\eta = (1-R)\left(1-e^{-\alpha d}\right)\xi$$

The engineering task is to maximize the efficiency.

Photodiode Characteristics: Efficiency

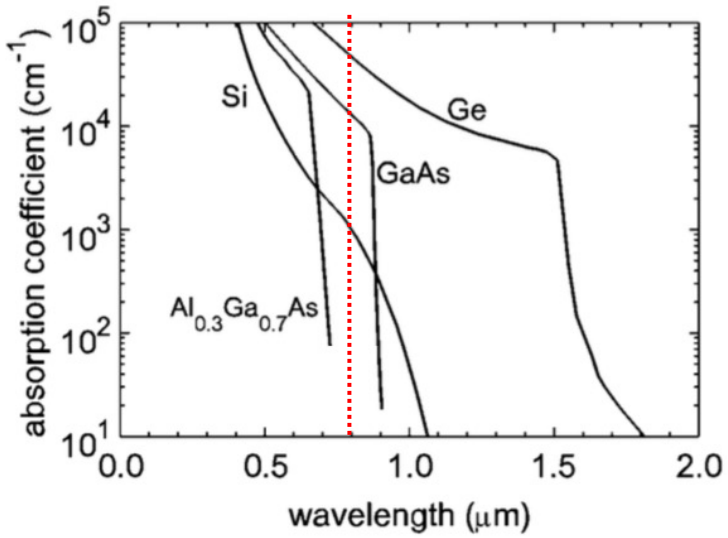
Quantum efficiency. The ratio of created ehp to number of incident photons.

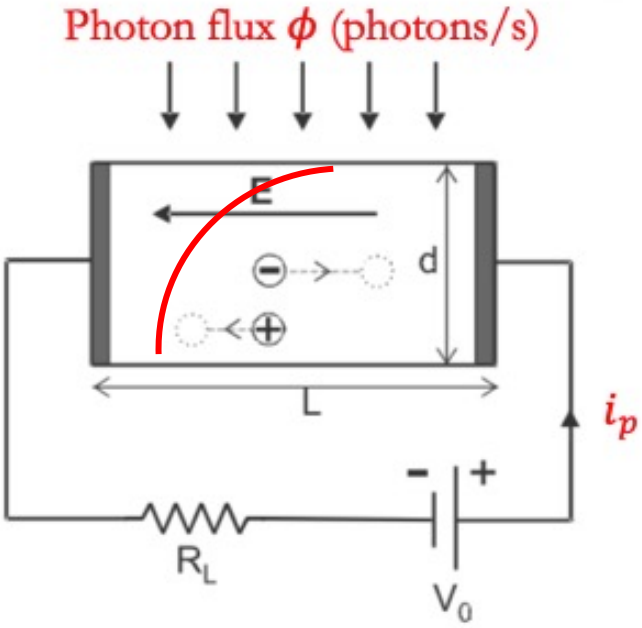
$$\eta = (1 - R)\left(1 - e^{-\alpha d}\right)\xi$$

ξ : Apply bias and clean fabrication process.

R : Use anti-reflection coating (small bandwidth)

αd : Choose best material and thickness.



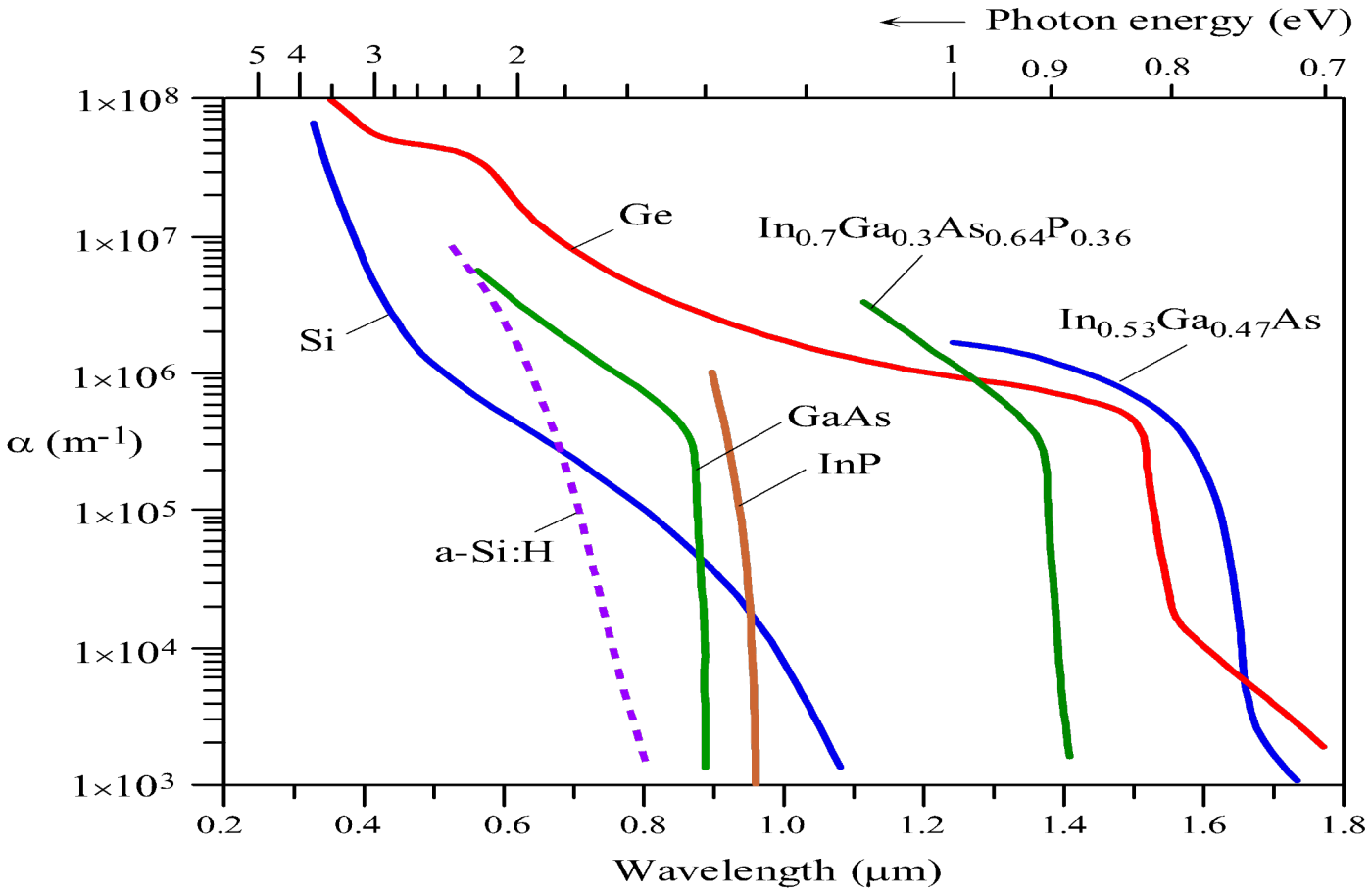


Example: For 95% absorption at 0.8 μm, find material and thickness.

$$1 - e^{-\alpha d} = 0.95$$

Photodiodes Characteristics: Efficiency

Absorption coefficient of common semiconductor materials.

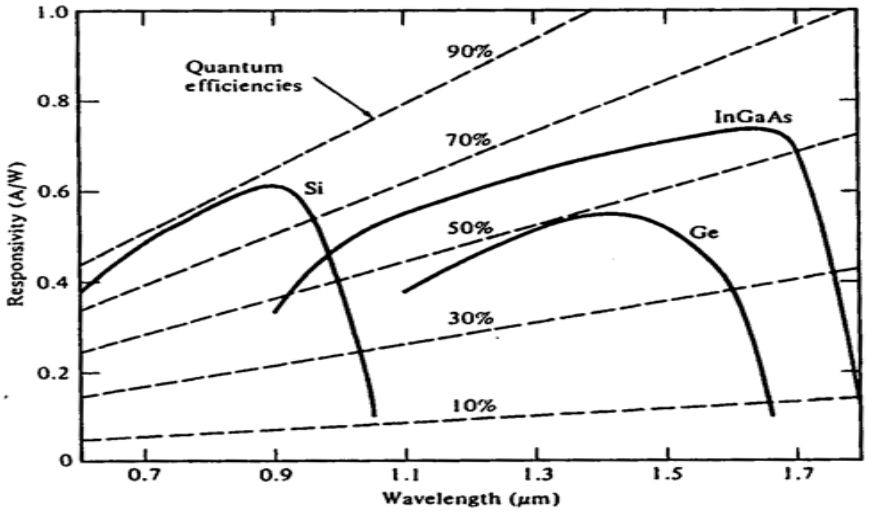


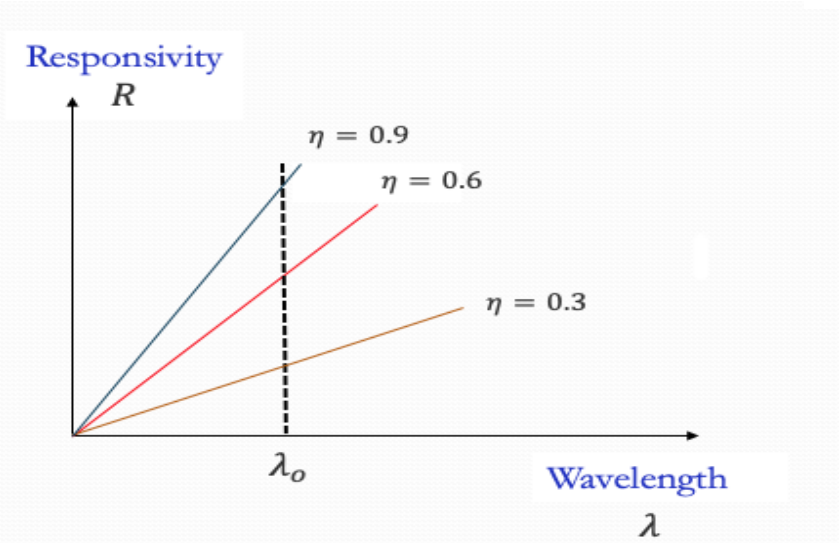
Photodiode Characteristics: Responsivity

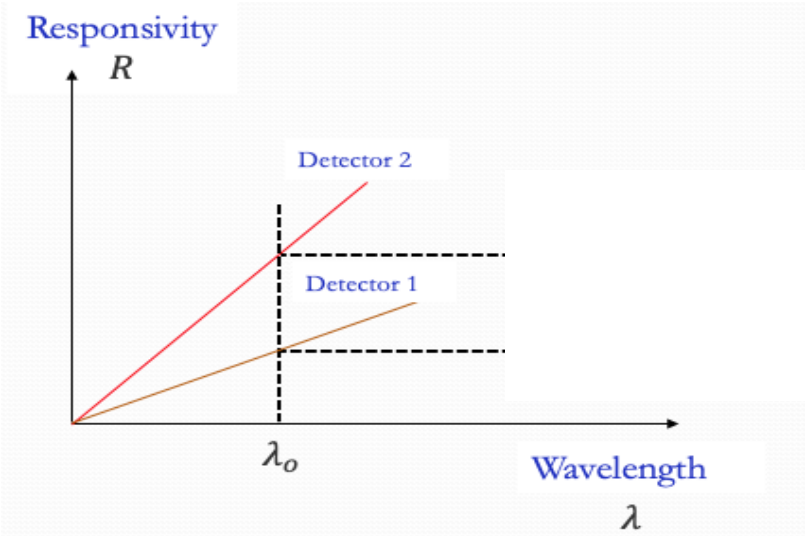
Responsivity. The ratio of optical current to incident optical power.

$$R = \frac{i_p}{P_o} = \eta \frac{e}{\hbar\omega} \qquad \text{(A/W)}$$

$$R = \frac{\eta e}{\hbar 2\pi c}\lambda$$

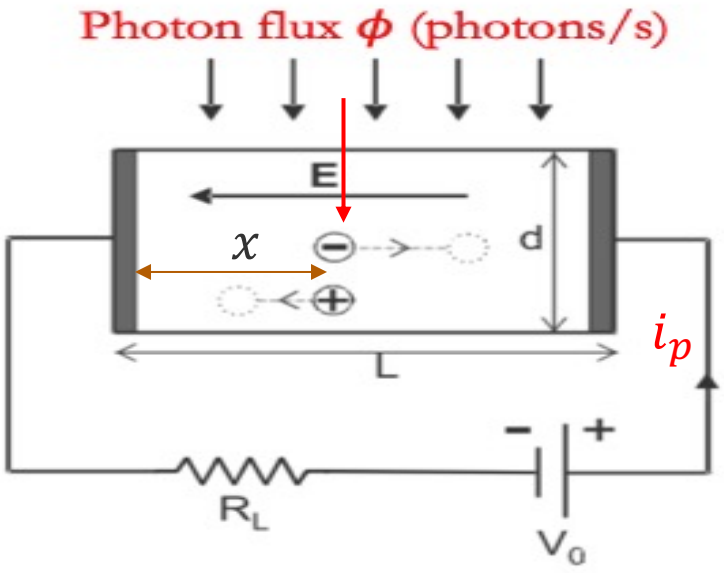


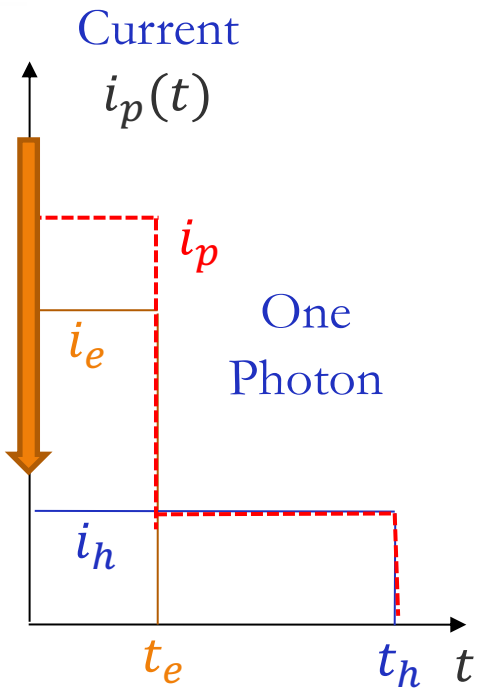


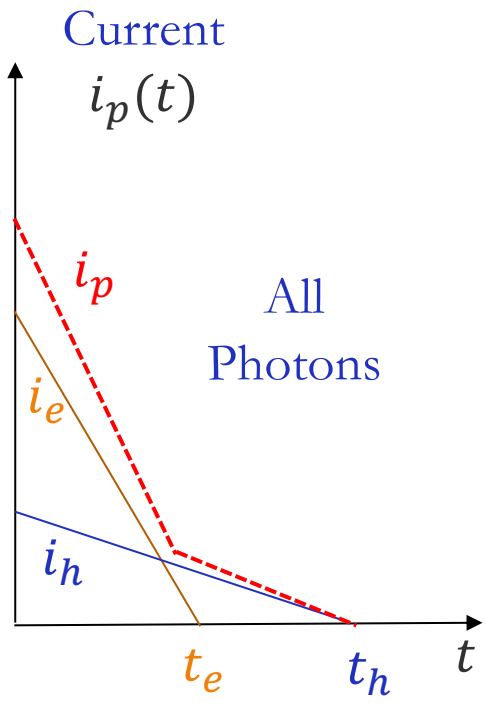


Photodiode Characteristics: Impulse Response

The impulse response is composed of transit time and RC delay.







$$i_e(t) = e\frac{v_e}{l}$$

$$i_h(t) = e\frac{v_h}{l}$$

$$i_p(t) = \frac{e}{l}(v_e + v_h)$$

$$t_e = \frac{L - x}{v_e}$$

$$t_h = \frac{x}{v_h}$$

Example. GaAs material with L= 1 cm, v_e=10^7 cm/s. i_e=1.6 pA.

Photodiode Characteristics: Impulse Response

The impulse response is composed of transit time and RC delay.

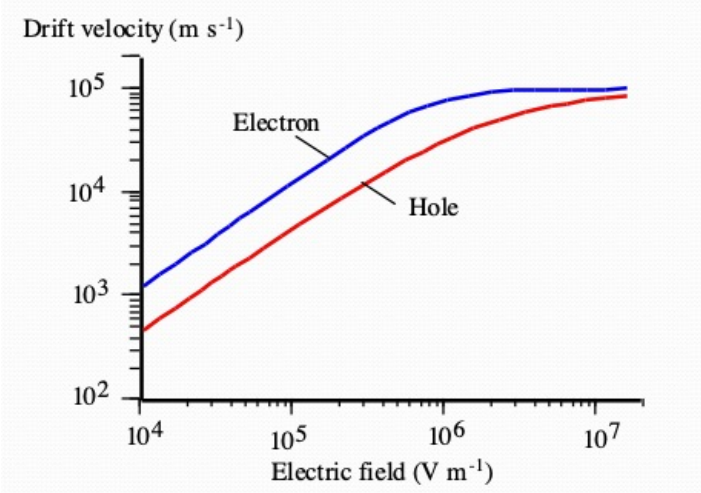
$$i_e(t) = e\frac{v_e}{L}$$

$$i_h(t) = e\frac{v_h}{L}$$

$$i_p(t) = \frac{e}{L}(v_e + v_h)$$

$$v_e = \mu_n E \qquad v_h = \mu_p E$$

cm²/Vs	μ_n	μ_p
Si	1500	450
Ge	3900	1900
InP	4600	150
InGaAs	14000	400



To improve the response:

- High mobility material.
- Small L.
- Drift velocity point.
- Help holes ($t_e = t_h$).

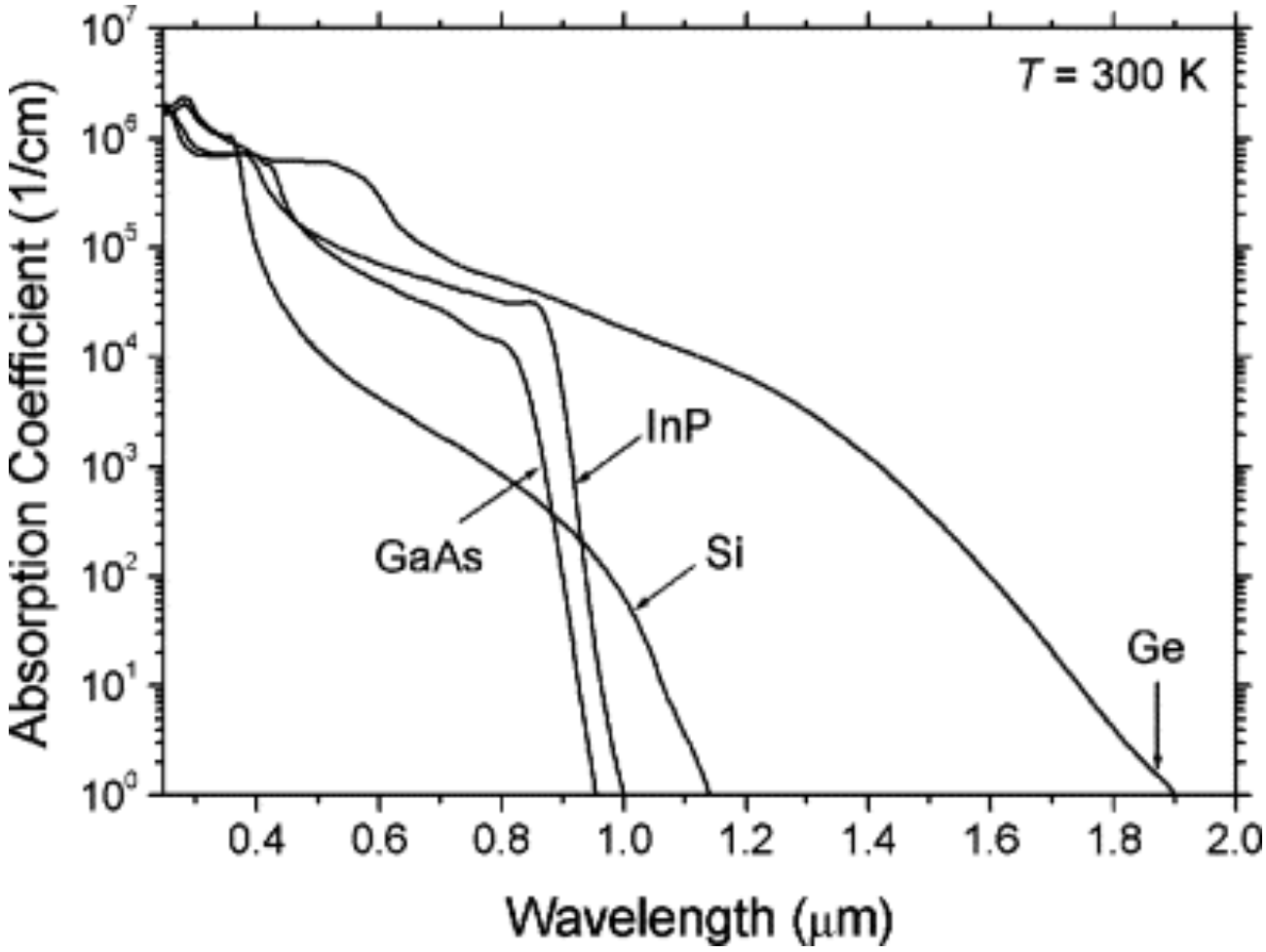
Cut-off Wavelength

$$\hbar\omega \geq E_g$$

$$\lambda_{co} = \frac{hc}{E_g} = \frac{1.24}{E_g}$$

	E_g(eV)	λ_{co}(μm)
Si	1.12	1.11
Ge	0.66	1.87
InP	1.35	0.91
InGaAs	0.75	1.65

Photodiode Characteristics

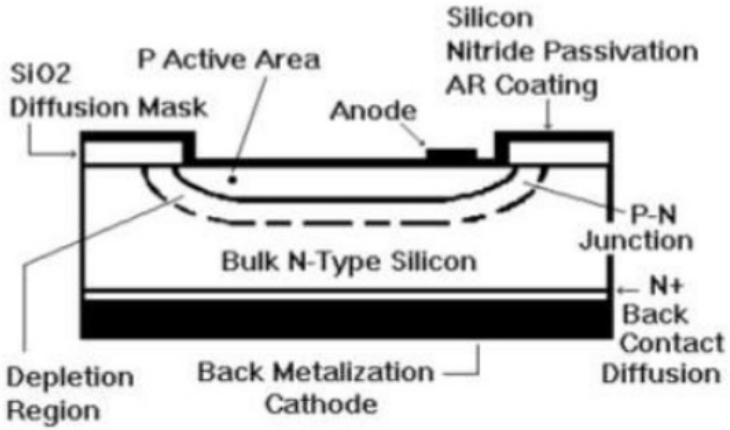


Photodiode Characteristics: Datasheet

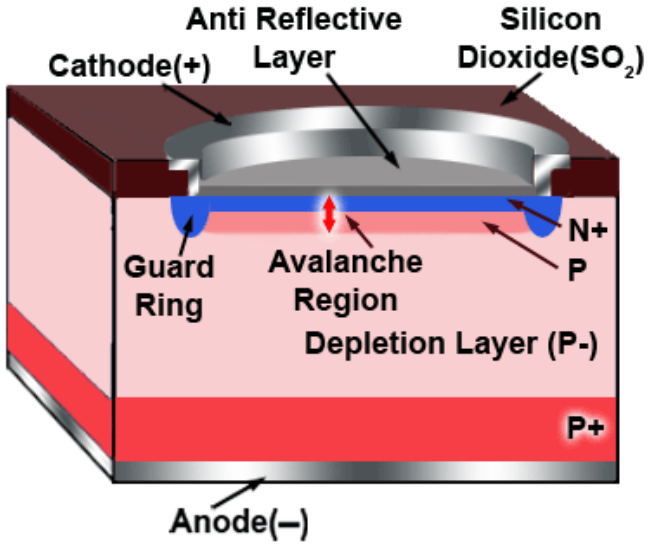
Parameter	Symbol	Unit	Si	Ge	InGaAs
Wavelength range	λ	nm	400-1100	800-1650	1100-1700
Responsivity	R	A/W	0.4-0.6	0.4 –0.5	0.75 – 0.95
Dark current	I_D	nA	1-10	50-500	0.5-2.0
Rise time	τ_r	ns	0.5-1	0.1-0.5	0.05-0.5
Bandwidth	B	GHz	0.3-0.7	0.5-3	1-2
Bias voltage	V_B	V	5	5-10	5

Photodiodes Types and Designs

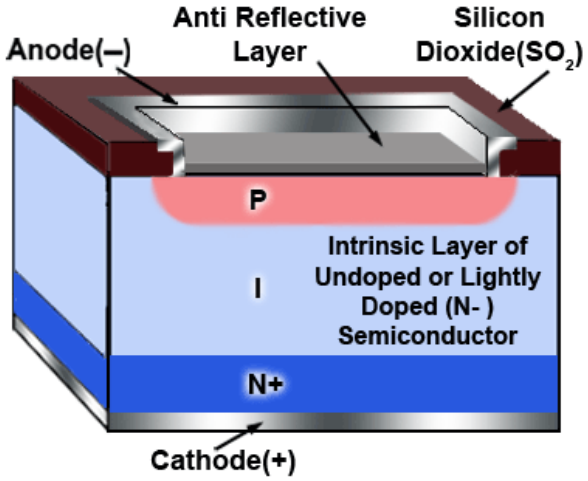
Basic PN photodiode.



Avalanche photodiode (APD).



PIN photodiode.





Examples

(1) A Si PIN with L=20 μm, V_{bias}=100V, λ=900 nm (pulsed). Discuss operation.

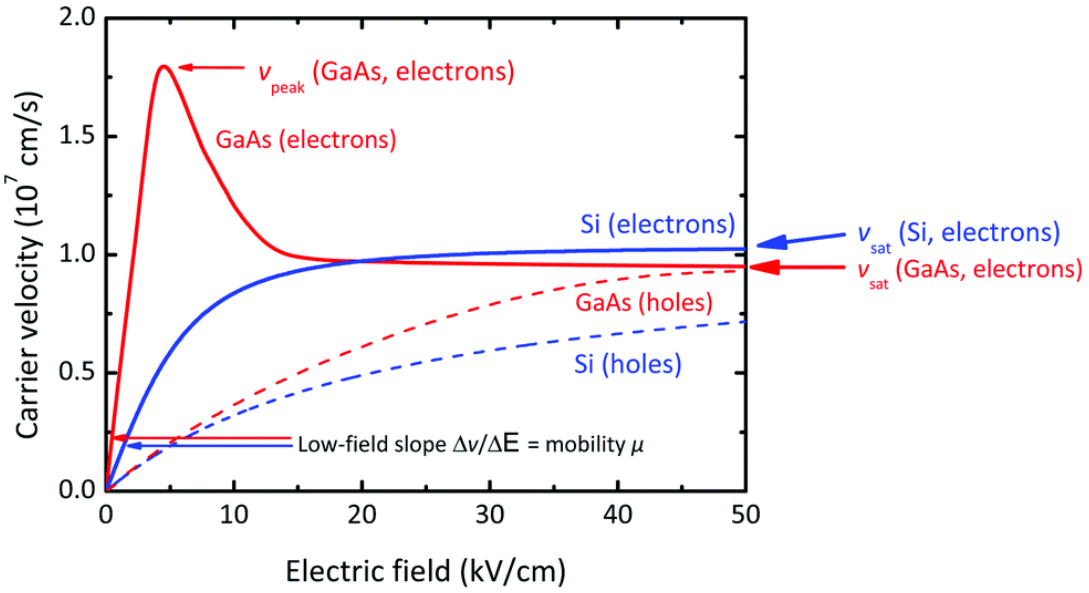
At 900 nm, $\alpha = 3\times10^4\ m^{-1}$

$$E = \frac{V_{bias}}{L} = 5\times10^6\ V/m$$

$$v_e = 10^5\ \ m/s$$

$$v_h = 7\times10^4\ \ m/s$$

$$t_h = \frac{L}{v_h} = 0.286\ ns$$



To improve the speed, decrease L, but that will hurt efficiency and hence responsivity. Find a trade-off.

Examples

(2) A Si PIN with diameter 0.4 mm is illuminated with a 700 nm light of intensity I=0.1 mW/m^2. The optical current produced was measured to be i_p=56.6 nA. Find efficiency and responsivity.

$$P_o = Area \times Intensity = A\,I = \pi(0.2\,mm)^2(0.1\,m)$$

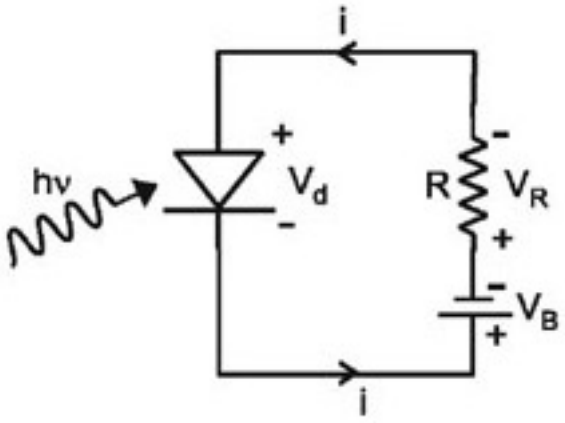
$$P_o = 1.26\times10^{-7}\,W$$

$$R = \frac{i_p}{P_o} = 0.45\,A/W$$

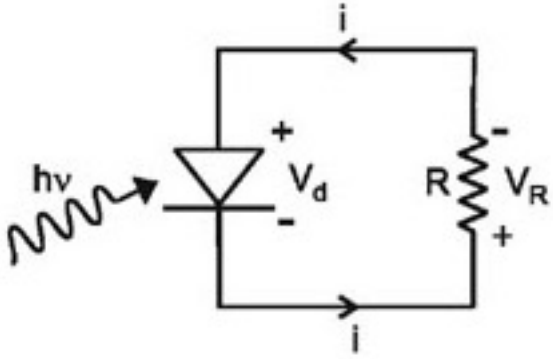
$$\eta = R\frac{hc}{e\lambda} = 80\%$$

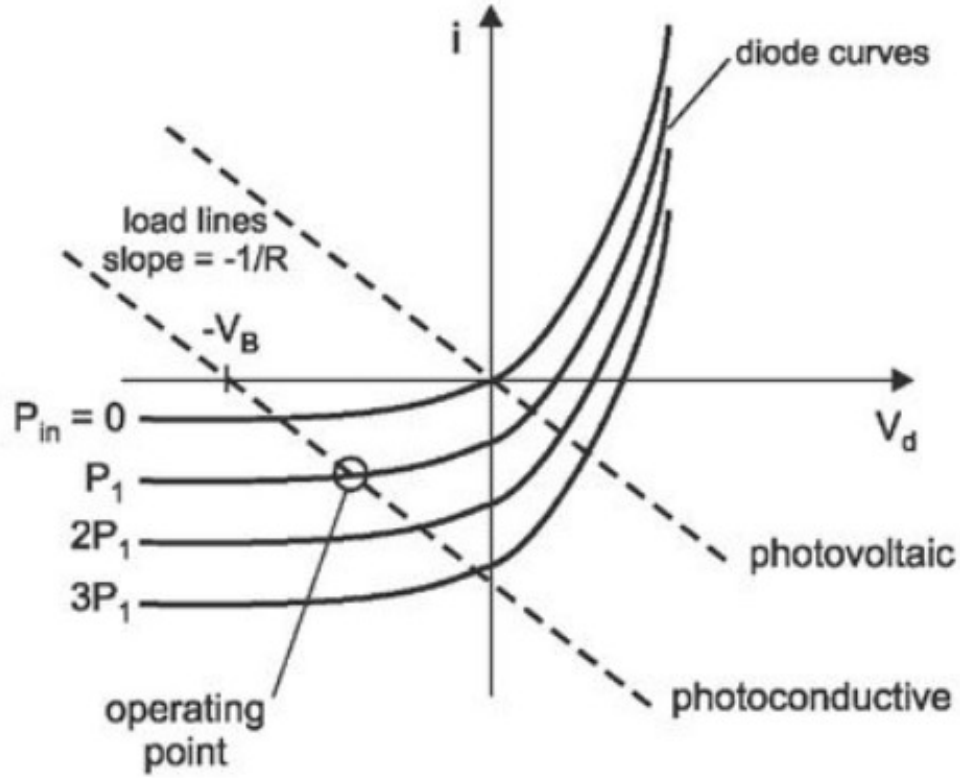
Photodiode Modes of Operation

Photoconductive Mode



Photovoltaic Mode





Q & A

Send me an email if you have any questions.